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SURVEY

Student Retention Using Educational Data Mining and Predictive Analytics: A Systematic Literature Review

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ABSTRACT Student retention is an essential measurement metric in education, indicated by retention rates, which are accumulated as students re-enroll from one academic year to the next. High retention rates can be obtained if institutions aim to provide appropriate support and teaching methods among the various practices to prevent students from deferring their studies. To address this pressing challenge faced by educational institutions, the underlying factors and the methodological aspects of building robust predictive models are reviewed and scrutinized. Educational Data Mining (EDM) and Learning Analytics (LA) have been widely adopted for knowledge discovery from educational data sources, improving the teaching practice, and identifying at-risk students. Various predictive techniques are applied in LA, such as Machine Learning (ML), Statistical Analysis, and Deep Learning (DL). To gain an in-depth review of these techniques, academic publications have been reviewed to highlight their potential to resolve Student Retention issues in education. Additionally, the paper presents a taxonomy of ML approaches and a comprehensive review of the success factors and the features that are not indicative of student performance in three different learning environments: Traditional Learning, Blended Learning, and Online Learning. The survey reveals that supervised ML and DL techniques are broadly applied in Student Retention. However, the application of ensemble and unsupervised learning clustering techniques supporting the heterogenous and homogenous groups of students is generally lacking. Moreover, static and traditional features are commonly used in student performance, ignoring vital factors such as educators-related, cognitive, and personal data. Furthermore, the paper highlights open challenges for future research directions.

INDEX TERMS Educational data mining, learning analytics, machine learning, predictive models, student retention.

I. INTRODUCTION

This section is divided into four major parts, emphasizing the context of this systematic literature review and the main research contributions. Firstly, we begin by introducing the main topic of this review, followed by the subtopics under the field of student retention.

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A. STUDENT RETENTION

Student retention is one of the pressing challenges in higher education. Retention rate is an essential measurement metric in education [1], [2]. As students re-enroll from one academic year to the next high retention rates are obtained [3], [4]. To achieve this, institutions should provide appropriate support and teaching methods [1] as among the various practices to enhance student performance [5] and prevent them from deferring their studies. This area has gained massive attention

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in the education sector over the years as many researchers and institutions are interested to learn about the factors preventing students from completing their degrees and what factors do not [6], [7].

Maintaining a high graduation rate in universities can be challenging due to poor student retention. General statistics show that the overall dropout rate for undergraduate college students is relatively high at 40%, with approximately 30% of freshmen discontinuing their studies before their second year [8], [9]. While the reasons for students choosing to drop out of universities vary widely based on the different views of student retention in institutions and different data used, the issue is still a major concern [10], especially for higher education institutions looking for enriched retention strategies. Many empirical and theoretical studies are focused on exploring the determinants that could lead students to drop out. One of the major factors identified by current studies is financial concerns [11], [12], due to costly education or low education funding.

Low student retention in universities can affect students financially [13]. As student discontinue their studies, a student's education loan must be paid even if the student fails to graduate, which leads to a low graduation rate in higher education institutions. As the demand increases for employing highly qualified students, failing to meet students' expectations can result in low-income jobs. Furthermore, low retention rates can be an indicator of the failure of institutions to accomplish their mission [14]. If a university fails to prepare young mindsets for the business world, this can lead to a massive loss in human capital [13] as fewer professionals and practical skills are produced by universities.

A high retention rate can bring many advantages to not just students but to educators and parents as well. For example, if the student is not active during the coursework activities prepared by lecturers, then an analysis of the student's progress can conclude practical points which help educators to refine their teaching methods, and materials for better student engagement and learning. A student that exhibits low progress or is likely to leave the university due to issues and other aspects are identified as ''at-risk'' students [14]. Prediction of these students as early as possible can bring many benefits and helpful tools to institutions and students, such as an early warning system [10], [15], [16] generated to resolve issues that might lead students to defer or discontinue their studies. Additionally, predicting the successful students can help educators identify the characteristics of an outstanding student, and these traits can be established as a set of principles that can be provided for students when they commence in a university.

Many institutions are turning to fine and high-tech solutions that are oriented toward data and analysis, known as Predictive Analytics, more specifically Learning Analytics. The purpose of employing analytics in education is to explore trends and patterns using numerous amounts of historical data to predict the future of students' success.

Many empirical studies have proved the significance of using Learning Analytics to enhance student retention [10], [14], [17], [18]. An institution such as Georgia State University and an estimate of 1,400 colleges and universities [19] are applying Learning Analytics to redirect students to a better and more successful career by using data to predict their overall results as early as possible before their second semester. Based on these predictions, recommendations are given to students, for example, suggestions to pursue specific programs, tutoring classes, and so on. Although student retention is generally related to ''student's degree completion'' however, this review did not solely focus on this goal as empirical studies pertaining to student retention and attrition in three different classroom environments such as online, traditional, and blended learning, are gathered and scrutinized.

B. EDUCATIONAL DATA MINING

Educational Data Mining (EDM) is an emerging research field applied in the education context [10]. It uses the various amounts of data obtained from institutions, including student data, administrative records, etc. It is vital to understand students' behaviours in educational institutions and identify successful and failed students to improve the teaching and learning environment. With the help of EDM, hidden patterns can be discovered, implying relationships between many attributes found in large datasets [20] obtained from university records.

EDM has been widely adopted in many educational applications [21], such as maintaining and improving courses, generating recommendations, creating alerts, detecting undesirable behaviours, and predicting performance. The term is extracted from ''Data Mining'', which is the process of searching through large datasets to identify valuable information. It extracts and analyses data to explore hidden patterns. Data Mining can be applied in many contexts such as business and financial analysis, healthcare, intrusion detection, and higher education. In Data Mining, we can obtain knowledge from the Knowledge Discovery in Databases (KDD) process [22], [23] following six main steps.

The process of KDD applies Data Mining techniques [24] to discover knowledge by:

1. Selecting specific data sources as the Target Data.
2. Cleaning the data to eliminate incomplete, inconsistent records to obtain a standardized dataset.
3. Transforming the data and applying Data Mining methods such as statistics or algorithms to discover previously unknown patterns.
4. Finally, the results are interpreted and evaluated to represent the discovered knowledge.

C. LEARNING ANALYTICS

Students interact with a university through various activities such as portal login, uploading assignments, class attendance, etc. often these actions leave behind digital traces that can be used to analyse how the university is managing

and or providing help for students. Such traces can be stored as potential data that is widely applicable in Learning Analytics (LA).

LA is defined as the process of analysing, measuring, collating, and reporting data [18] to make decisions about the progress made by students and educators. Student retention is an issue that most universities are currently dealing with [25], [26]. Educators might face challenges in retaining a student until the end of their academic year as there can be several factors affecting the education of the student. For this purpose, LA has been introduced to efficiently analyse the learning of the students that are affected by factors such as demographics, interaction through clickstream behaviour, and so on [27].

Therefore, LA techniques can be adopted to identify the factors related to student dropout and thus, prepare a better study environment to ensure students are able to earn their degree on time. Learning Analytics and Educational Data Mining both aim for the same goal, which is to improve the quality of student management and their learning environment. In a recent review by Lemay *et al.* in 2021 [28], it is revealed that there's an increasing overlap between these two terms as they have been interchangeably used and therefore results in an ongoing confusion [29] in the field of student retention. Thus, we summarized the findings based on three (3) recent survey articles ([28], [30], [31]) exploring the difference between LA and EDM as depicted in Table 1.

TABLE 1. Similarities and distinctions of learning analytics and educational data mining.

	Learning Analytics	Educational Data Mining
Topics/ Focus [28]	Student engagement, teaching tools, and social network analysis	Techniques and methods of data analysis (e.g., classification)
	The processes of learning of individuals	Knowledge discovery from other data sources
	Applications to teaching practice	Performance and system each year
	Students' performance or learning in a course	Overall performance; not for a specific course
Top keywords [28]	Online, social, systems, collaborative, and student.	Performance, prediction, analysis, machine, classification, and student.
Major research themes [31]	Student-focused learning objectives	Algorithms behind student performance prediction
Similarities [28], [30]	Focus on student performance and learning platforms, and modelling student behavior.	
	Frequently conducted in the same research settings (e.g., virtual learning environment).	

We, therefore, conclude that both terms should be discussed separately, although they result in the same purpose and goals, however, LA is targeted towards specific courses in institutions to analysing student's performance and behaviour whereas EDM focuses on the technical aspect and methods used to examine student performance more generally.

D. RESEARCH HIGHLIGHT & CONTRIBUTION

This systematic literature review (SLR) provides an extensive analysis of 100 articles on existing techniques using Learning Analytics to further improve education and the learning environment. Our contribution to the study can be summarized as follows:

1) Unlike other review papers, this SLR also highlights the unsuccessful student factors that fail to predict and understand the behaviour of students. This will ease the feature selection process and improve the accuracy of predictive models.
2) A comparative study of existing literature along with their strengths and limitations to induce a valuable research gap.
3) The existing work is reviewed and summarized according to three learning environments: traditional, online, and blended to aid future researchers in understanding which features are suited for a specific teaching and learning environment.
4) A taxonomy of current Machine Learning and Deep Learning algorithms used for predictive models to highlight the most applied algorithms in LA.
5) A review that focuses on multiple paper types, such as journal, conference, thesis, and case studies from a total of 14 bibliographic sources to include findings from different perspectives.

The rest of the paper is organized as follows. Section II covers the related work where other review papers will be compared in detail. Section III explains the methodology adopted to perform this systematic literature review (SLR) whereby the collection sources, keywords, model, and assessment criteria are presented. Section IV includes the results of the surveyed articles with visuals along with valuable key findings extracted to resolve retention problems with Learning Analytics. Section V presents the results of this review, where the formulated research questions are answered with a discussion of the obtained results to induce the research gap. In section VI, our research is concluded to review the importance and content of this systematic review paper, the limitations, and finally, the suggestions for further work of this review paper are provided.

II. RELATED WORK

This section explains the existing review papers and their limitations are summarized and compared.

There has been an extensive number of published reviews regarding the prediction of student performance and its relevance. The authors have reviewed ten (10) different types of review papers in this field of EDM. The review articles are distributed over the past five years (2017 to 2021) as can be seen in the comparative analysis in Table 2. We have reviewed different literature surveys in the field of student performance prediction. It has been concluded that no review was carried out to highlight the features that are not indicative of student performance in predicting student performance. To the best of

TABLE 2. Comparative analysis of existing review articles.

#	Ref. & Publication Year	Focus/ Scope of Survey	Type of Review	No. of Databases Explored	No. of Reviewed Papers; Their Types	Years Covered	Strengths	Limitation(s)
	Current Review **(2022)**	Educational Data Mining and Predictive Analytics in three learning environments	Systematic Review	14	100; Conference, Journal Articles, Case Studies & Thesis	(2017-2021)	Identifies the success factors and the features that are not indicative of predicting the student's performance; Reviews the current literature in line with three learning environments.	Refer to Section VI (B).
1.	[5] **(2021)**	ML techniques to predict student performance	Systematic Review	6	78; Conference & Journal Papers	(2009-2021)	Well formulated questions and discuss remedial action plans.	No comparison of the reviewed articles.
2.	[40] **(2021)**	Data Mining and LA techniques to predict student performance	Systematic Review	7	64; Conference & Journal Papers	(2010-2020)	A comprehensive review with practical limitations and recommendations.	No information regarding the factors used in prediction.
3.	[39] **(2020)**	Prediction of academic success in higher education	Traditional Review	7	N/A; N/A	(2016-2020)	Provides a holistic view and review of applying predictive techniques.	Only presents the influential factors.
4.	[38] **(2020)**	ML techniques to analyze & predict student performance	Traditional Review	14	64; Conference, Journal Papers & Book Chapters	(2015-2020)	Provides a summarized table of the main features of the reviewed articles.	Does not include details about data collection methods or factors used.
5.	[32] **(2019)**	Predictive Analysis Tools in Higher Education	Systematic Review	10	26; Conference, Journal Papers, & Book Extracts	(2012-2019)	Analyzed articles focusing on more than one prediction goal.	A limited number of papers were reviewed.
6.	[37] **(2019)**	Predictive modeling technique for monitoring student academic performance	Traditional Review	N/A	54; N/A	N/A	Reviewed the literature according to different applications of student performance.	No comparative analysis of articles.
7.	[36] **(2018)**	ML techniques for MOOC dropout Prediction	Traditional Review	N/A	N/A; N/A	N/A	Provides solutions to tackle student dropout issues	Does not include details about data collection methods.
8.	[35] **(2018)**	Predicting Academic Performance	Systematic Review	3	497; Journal Papers	(2010-2017)	Explains limitations and issues from the quality of predictive techniques	Does not identify features that are not indicative of student performance.
9.	[33] **(2017)**	Using data mining for student performance prediction	Literature Survey	6	24; Conference, Journal, Workshop Papers, Expert Lectures/Talks, Blogs, Communities	(2007-2016)	Presents predictive data mining techniques focus on successful factors	A limited number of papers reviewed; doesn't include limitations of reviewed articles.
10.	[34] **(2017)**	Prediction of student learning outcomes using LA	Systematic Review	N/A	39; Conference & Journal Papers	(2002-2016)	Classifies predictive models and articles according to accuracies and methodologies	a preliminary study is done which lacks the LA techniques used and other important details.

** (N/A) denotes unavailable information; Based on the author's knowledge.*

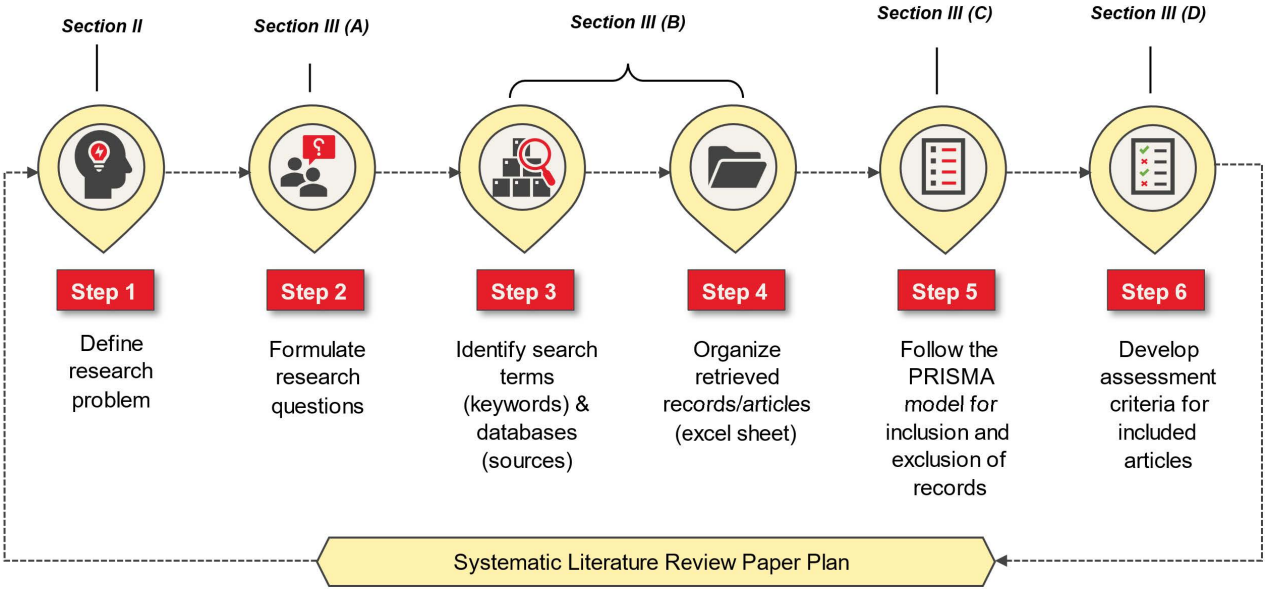


FIGURE 1. Systematic literature review paper plan.

our knowledge, existing reviews focus on factors contributing to the student performance predictions and ignore factors that do not work. In Table 1, existing surveys in the field of learning analytics and the prediction of student performance are summarized to highlight their strengths and weaknesses. While these existing surveys present prominent findings and interesting future directions, they lack significant features of review such as:

1) A Limited number of reviewed articles [32]–[34], which makes it harder to gain an in-depth understanding of the current methods used and the limitations.
2) Most surveys focus on identifying the successful factors ignoring the less significant factors [5], [32]–[40].
3) Quality is unsatisfactory [36], [37] (for example, lacking the methodologies used, no taxonomy presented).
4) Does not include a comparative analysis of articles [5], [34]–[37], [40]; this is an indispensable element for review papers since it provides deep analysis and in turn, it will be easier for readers to identify potential research gaps and areas to improve on.

This systematic literature review (SLR) aims to provide useful insights to forthcoming researchers looking to resolve student retention using learning analytics and predictive models. The paper focuses on introducing the features that significantly contribute to the prediction of student performance as well as the less significant features to ease the feature selection process. Aside from that, the reviewed articles will be extensively analysed and compared to highlight their strengths and limitations.

III. SURVEY METHODOLOGY

This section introduces a thorough understanding of the research methodology to explain how this SLR was performed. Compared to a traditional review, a systematic review is an evidence-based analysis of relevant studies. A narrative review may result in biases and poor review quality as authors select articles manually [41]. Therefore, we chose a systematic review to summarize evidence-based on formulated questions with a detailed and comprehensive plan [42] as illustrated in Figure 1. The authors have prepared and followed a 6-steps plan to execute the review rigorously and avoid biases. This review paper plan includes the organization of information for this section and the important steps to reach a precise conclusion to address the identified research problem explained in section I (A). Figure 1 presents the detailed flowchart of the review plan.

A. DATA COLLECTION & EXTRACTION

Firstly, the authors formulated three main questions to help in searching and identifying the correct studies to tackle the issue of student retention and forecasting the students' performance. The literature review has been carried out to provide elaborative answers to the following three (3) key research questions formulated below:

- ***Research Question 1 (RQ1):*** Student performance factors: (A) What kind of data sources and factors can help us to determine high-risk students and (B) what factors fail to forecast the performance and behaviour of students?
- ***Research Question 2 (RQ2):*** Learning Analytics Approaches: What learning analytics approaches can be used to improve the student's performance and teaching practice?
- ***Research Question 3 (RQ3):*** Machine Learning-based Approaches: What kind of algorithms are mostly used by researchers to develop predictive models in student retention with high accuracy?

TABLE 3. Databases and sources used.

No.	Database	URL	Retrieved Papers
1.	Google Scholar	https://scholar.google.com/	71
2.	IEEE Xplore Digital Library	https://ieeexplore.ieee.org/	48
3.	Elsevier (ScienceDirect)	https://www.sciencedirect.com/	29
4.	Springer (SpringerLink)	https://link.springer.com/	23
5.	ResearchGate	https://www.researchgate.net/	24
6.	MDPI	https://www.mdpi.com/	21
7.	Taylor & Francis	https://www.tandfonline.com/	17
8.	Semantic Scholar	https://www.semanticscholar.org/	16
9.	ACM Library	https://dl.acm.org/	13
10.	Emerald Insight	https://www.emerald.com/insight/	12
11.	IOPscience	https://iopscience.iop.org/	10
12.	Wiley	https://www.wiley.com/en-us	9

B. KEYWORDS & SOURCES USED

The SLR carried out an extensive searching process based on the objectives of the research questions in this review. The authors exploited 14 research sources in total to find the relevant and recent articles. The top 12 bibliographic databases with the number of retrieved papers are listed in Table 3. Other sources included Sage (4 retrieved papers) and ProQuest (3 retrieved papers).

The repositories were explored using different queries to find data and papers related to the research questions. The search was filtered to restrict the publication year between 2017 and 2021. The literature research began in early 2021, therefore, the authors considered reviewing articles up until this year. We have chosen to select papers from the past five years to ensure only relevant and up-to-date findings are revealed by this survey. Articles that did not meet the query below are disregarded to make sure only the most relevant papers are retrieved. However, since Google Scholar is a famous indexer of various sources, there were few duplicates with other databases. These duplicates were removed, as explained in the following section.

The authors have mainly used Microsoft Excel as a tool for organizing, listing, and sorting the retrieved papers. Duplicates were removed after the sortation of the article title was performed. Statistics and figures were obtained through the PivotTable Analyze tool in excel for the analysis of articles according to publication year, paper type, etc. Furthermore, in the excel sheet, we marked an article as ''inclusion or exclusion'' based on the author's decision to make sure only the relevant articles are surveyed from the list. The decision is made based on the criteria depicted in Table 4.

Prior to conducting this SLR, the authors have performed a preliminary search and gathered 75 articles to gain an understating of the domain and common keywords used by researchers in the LA field. The keywords were extracted, and a count was obtained for each term out of 161 total keywords. To justify our selection of the keywords used in the search queries below, we concluded essential keywords that have appeared more than twice:

1) ***"EDM"*** appeared 17 times.
2) ***"Learning Analytics"*** and ***"Student Performance"*** appeared 14 times each.
3) ***"Student Retention"*** and ***"Predictive models"*** appeared four times each.
4) ***"Dropout prediction"*** appeared three times.

We have excluded specific keywords that indicate the name of techniques, for example, ''classification'', ''decision tree'', ''classification'' and similar terms. This is because this review aims to report a holistic picture of different techniques and algorithms used to address student retention. Similarly, for the sample population, the search query was not catered to a specific educational course or institution name. other keywords in the query, such as ''Predictive algorithm'' and ''at-risk student'' were chosen from other relevant SLRs [5], [32].

Below are the search terms used to extract relevant papers from the databases according to our research questions:

- **Techniques**: (''EDM'' OR ''Predictive Models'' OR ''Predictive Algorithm'' OR ''Learning Analytics'')
- **Purpose**: (AND ''Student Performance'' OR ''Dropout Prediction'' OR ''At-risk student'' OR ''Student Retention'')
- **Sample Population**: (AND ''Education'' OR ''University'' OR ''School'' OR ''Postgraduate'')

C. PRISMA MODEL

The research ensured a superior methodology to be followed based on the latest PRISMA statement and reporting guidelines presented by authors Page *et al.*, [43] as of 2020. The updated PRISMA flow diagram indicates the importance of including a summary of the previous related reviews. To emphasize this, the authors have conducted a comparative analysis of the existing review articles presented in section II. Additionally, we have used the PRISMA model to report the step-by-step procedures followed in searching and identifying of the articles used in this review.

As depicted in Figure 2, the PRISMA process begins with the identification phase whereby the total number of records identified and retrieved is reported. A total of 300 articles are retrieved. Later, 25 duplicate records are removed, leading to the next phase, screening. In the screening phase, the records are excluded if they're not related to the topic of this review and if they are not in the right range of the publication year. Next, the articles are assessed for eligibility for inclusion according to the list presented in Table 4, resulting in the inclusion of 100 papers in the final stage. These papers are further analysed and discussed in the following sections. The search is executed to obtain the list of

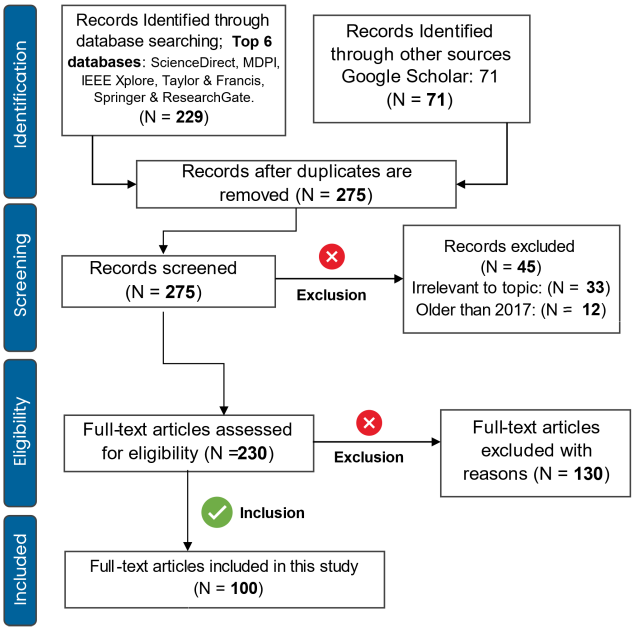


FIGURE 2. PRISMA flow diagram.

studies that can be used for further evaluation. These studies are managed using a renowned bibliography tool known as "Mendeley v1.19.8".

D. ASSESSMENT CRITERIA

To induce a research gap and highlight the limitations of the current literature, we identified and followed eight (8) main criteria as listed below (the results and assessment of these criteria are summarized and explained in section IV (D)):

- **Criteria 1 (C01):** Data collection methods, attributes, and dataset details are clearly stated.
- **Criteria 2 (C02):** Authors address and resolve the ethical issues of conducting the study.
- **Criteria 3 (C03):** Models and algorithms address data imbalance and overfitting issues.
- **Criteria 4 (C04):** Authors apply Feature Engineering
- **Criteria 5 (C05):** The predictor variables considered are clearly identified and described
- **Criteria 6 (C06):** Authors make use of separate (split dataset) training and testing datasets.
- **Criteria 7 (C07):** The predictive model is tested on an unseen dataset or verified with a second dataset.
- **Criteria 8 (C08):** The sample or dataset size includes more than 1000 instances.

IV. RESULTS & KEY FINDINGS

In this section, the result of surveyed articles is reported. The reviewed articles are distributed based on general information, dataset, sample, collection methods, learning type, algorithms, used languages, and tools. Lastly, the outcome of the assessment criteria is presented and explained.

TABLE 4. Inclusion and exclusion criteria.

No.	Inclusion criteria	Exclusion criteria
1.	Using Learning Analytics to predict at-risk students & their performance	Studies that do not relate to predictive learning analytics and student performance
2.	Articles published between the 2017 to 2021 era	Articles older than 2017
3.	Journal Articles	Review Articles
4.	Conference Papers	Book Chapters
5.	Case Studies	Working Papers/ Preprints
6.	Thesis / Dissertation	Articles not written in the English language.

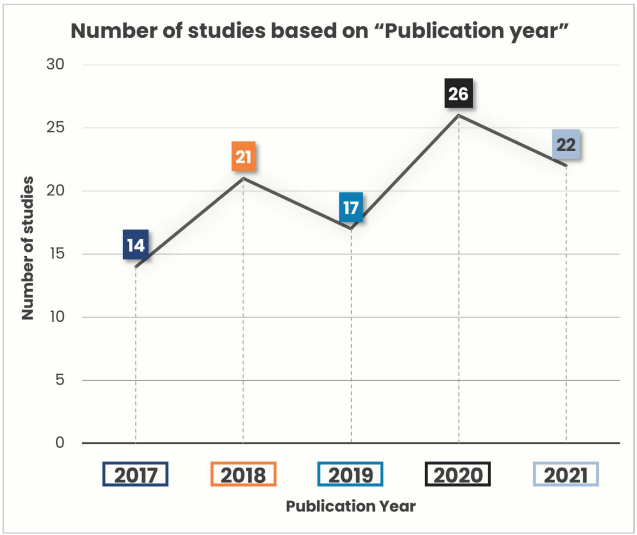


FIGURE 3. Number of studies published in the year 2017 to 2021.

A. GENERAL INFORMATION

General information includes the number of articles in the year 2017 to 2021 and the type of surveyed articles. The authors surveyed and analysed 100 studies to address the questions explained in section III (A). In Figure 3, a line chart is presented to depict the distribution of articles based on each publication year. The figure represents the total number of studies per year. For example, the highest number of surveyed studies is in the year 2020 (26 studies) and the second highest is in the year 2021 (22 studies). This review ensured a fair distribution in the number of articles per year to induce accurate findings and valuable research gaps for future researchers to consider in this field. Furthermore, we can see from the Figure that in the year 2017, the number of surveyed studies was 14, which indicates that the topic of learning analytics in education is constantly on the rise among researchers, and more studies will be presented in the future for further enhancement of learning methods in education.

Whereas, for the type of surveyed articles, this review focused on journal papers (55 studies) and conference papers (27 studies) as the two main types presented in Table 5. Other types include thesis and case studies. Unlike most current

TABLE 5. Types of analyzed articles.

Paper Type	No. of Studies	References
Journal Articles	55	[10], [13], [15], [16], [27], [44]–[93]
Conference Papers	27	[20], [23], [94]–[118]
Thesis/Dissertations	10	[119]–[128]
Case Studies	8	[129]–[136]

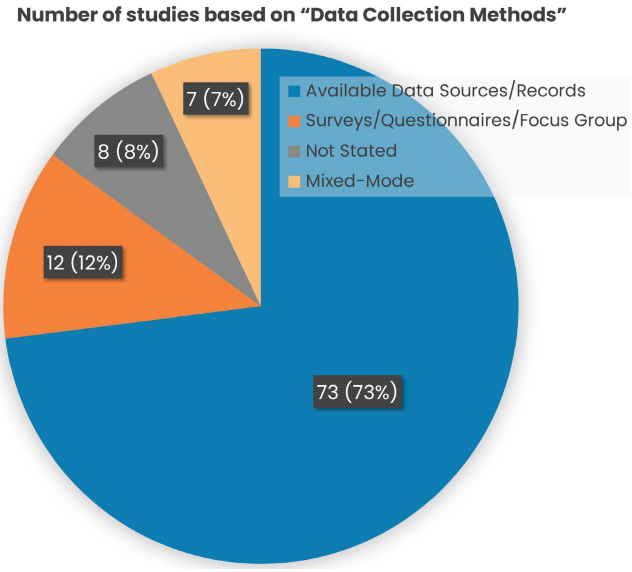


FIGURE 4. Number of studies based on data collection methods.

review papers that focus mainly on the journal and conference articles evident from Table 2, this review included case studies and thesis as it helps to look at the student retention problem from various perspectives. The authors ensured the inclusion of articles that present a sound methodology and comprehensive experimental results of the predictive models.

B. DATASET, SAMPLE POPULATION, SIZE & COLLECTION METHODS

This paper aims to provide a comprehensive review targeting important stages in developing predictive models for the predicting and understanding student performance or behaviours. Before the application of learning approaches, such as predictive algorithms, a crucial step takes place, which is the data collection and processing. Handling and producing high-quality data are often known as the most challenging task in predictive learning analytics. Therefore, reducing the number of features to include only the influential factors can highly improve the prediction accuracy [104], [137].

In this step, we analysed the articles based on the quality of the dataset, in terms of the features used, the sample size, population, and collection methods. As can be seen in Figure 4, majority of researchers have used available data sources and records found in public dataset or through university/school repositories, precisely

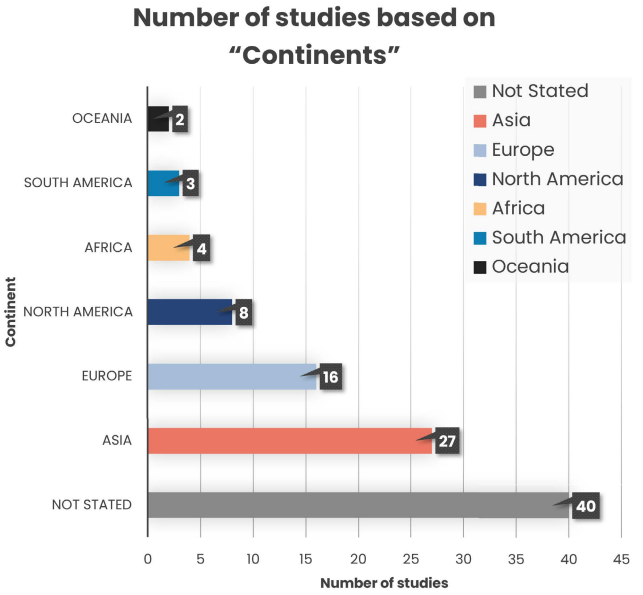


FIGURE 5. Number of studies based on continents.

a total of seventy-three (73) studies indicate this ([10], [13], [56], [58]–[66], [20], [69], [71]–[78], [81], [27], [82], [83], [85], [87]–[93], [45], [94]–[97], [100]–[104], [106], [49], [107]–[110], [113]–[117], [119], [50], [121], [123]–[126], [128], [129], [131], [132], [134], [51], [135], [136], [138], [54], [55]). While these sources are easily accessible and provide expected accuracy of the prediction models, some researchers opted for a mixed-mode of data collection where both available sources and questionnaires data are utilized, precisely seven (7) studies indicated this ([49], [52], [79], [80], [84], [86], [120]). However, in some cases, the use of additional data collection methods may introduce errors in sampling issues and therefore leads to a decrease in the overall prediction accuracy, therefore it is observed that questionnaire data does not improve prediction unless feature selection is performed appropriately [79]. Lastly, twelve (12) studies have opted solely for surveys or questionnaires, and focus group for data collection ([23], [47], [122], [133], [48], [52], [53], [61], [67], [79], [98], [120]). Moreover, 8 studies have not explicitly discussed the data collection methods.

In Figure 5, we observe, firstly, the location of where the study took place. It is safe to say that Learning Analytics is widely adopted in institutions worldwide. With a total of fifteen (15) studies conducted in major developed countries such as (The United Kingdom, The United States of America, Canada, and Germany) ([27], [50], [102], [119], [123], [125], [127], [58], [59], [64], [65], [69], [89], [92], [96]). While the rest of the studies are conducted in developing countries, such as (Malaysia, Bangladesh, Indonesia, and so on), precisely forty-four (44) studies. The figure also concludes that Asia is among the leading continent in this field with 27 studies conducted in LA and the second-leading continent is Europe with 16 studies. Moreover, 40 studies have not stated the location of where the study was conducted, as can be seen from the Figure.

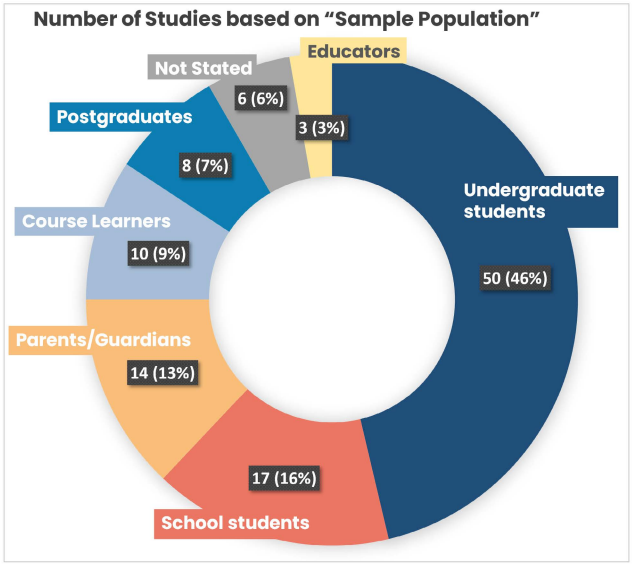


FIGURE 6. Number of studies based on the sample population.

Figure 6 highlights the distribution of the reviewed studies according to the sample population used. While most researchers took student-centred approaches, with nearly half of the studies (47%) focusing on undergraduate students, very few considered the inclusion of educators/teachers in their predictive models, precisely three (3) articles mentioned this [48], [95], [118]. These findings introduce a possible research gap for future researchers to develop a more comprehensive prediction model. It is worth noting here that the depicted numbers add up to more than 100 as researchers can build predictive models for postgraduate students along with educators, for example, or for both graduates and undergraduate and so on.

Figure 7 presents a bar of the frequency of studies based on the sample size. It is observed that only twenty-six (26) studies used a dataset with more than or equal to 1000 instances to train the predictive models ([13], [16], [58], [65], [68], [69], [80], [89], [90], [99], [107], [112], [27], [119]–[121], [123], [127], [129], [45], [46], [50], [51], [53], [55], [56]). However, it is known that most Machine Learning algorithms require a massive amount of data to perform well and produce accurate prediction results. These findings again introduce a possible research gap. Lastly, it is worth mentioning here that the statistics in the figure also represent the number of studies utilizing more than one dataset with different sample sizes, therefore, the total number of studies adds up to 104.

C. PREDICTION APPROACHES, ALGORITHMS, LEARNING OUTCOME TYPE & TOOLS

This subsection will explain the next stage of building student performance predictive models will be explained. It is divided into four (4) main parts, prediction approaches used, the algorithms, specifically in the ML field, the learning outcome, the types of the predictor outcome variable, and finally, the tools used. This review also summarizes publicly available student performance datasets presented in Table 7.

Figure 8 summarizes the reviewed articles according to the Learning Analytics approach implemented to forecast student performance. It is worth noting here that the depicted numbers add up to more than 100 as researchers may apply a combination of approaches such as statistical techniques

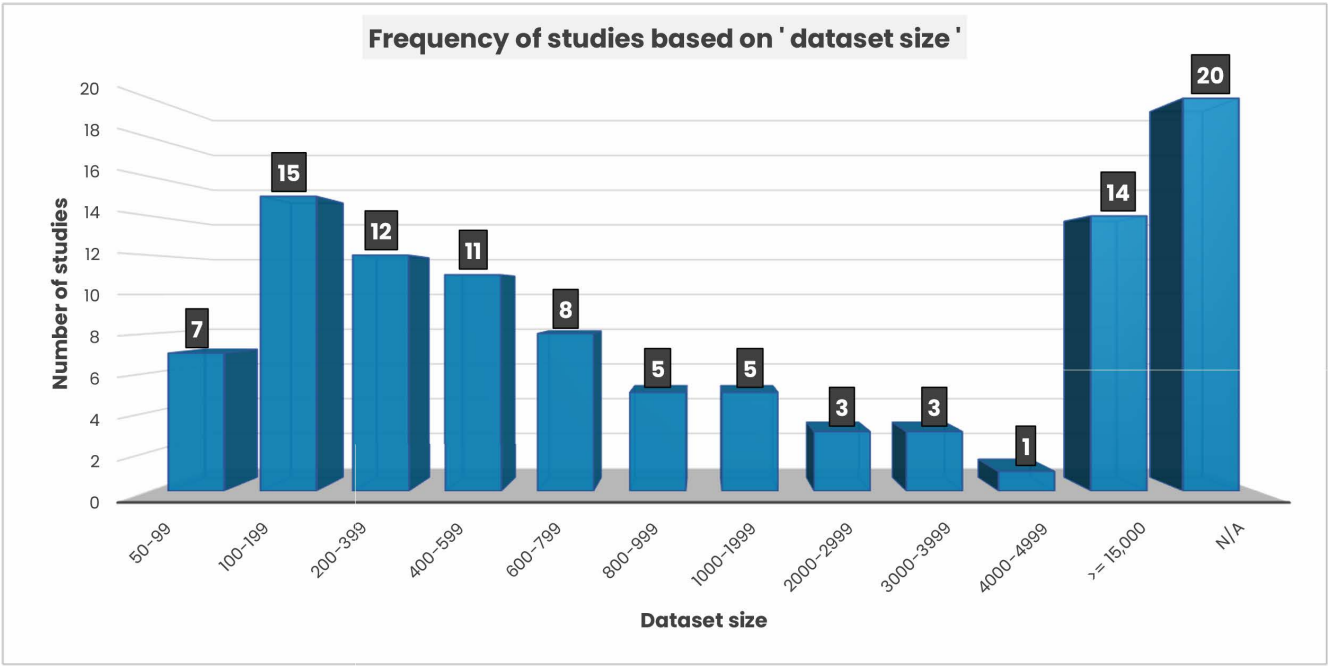


FIGURE 7. Number of Studies based on sample dataset size.

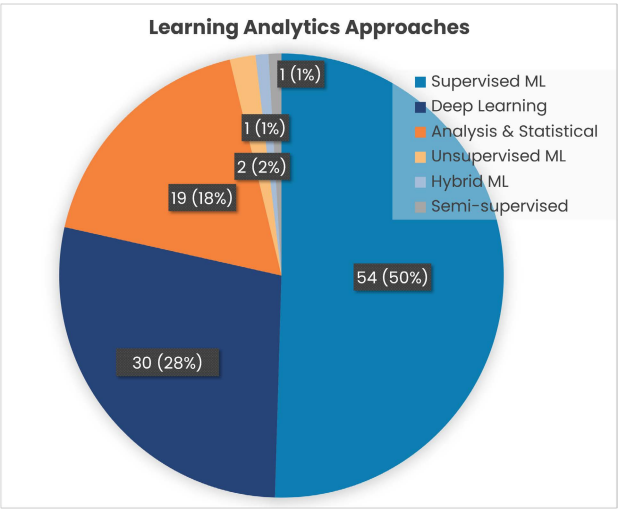


FIGURE 8. Number of studies based on predictive approaches.

along with supervised or deep learning and so on. Overall, there are Six (6) intelligent approaches are extracted from this review. It is observed that almost half of the studies have applied Supervised Machine Learning approaches (54 studies, 50%), then followed by Deep Learning approaches (30 studies, 28%), lastly in the top 3 approaches, are Analysis and Statistical approaches (19 studies, 18%). However, our findings show that very few researchers explored other worthy approaches such as Unsupervised Learning (2 studies (as seen in [47], [85]), 2%), Hybrid Machine Learning (1 study, 1% (as seen in ([73])), and Semi-supervised (1 study, 1% (as seen in [111])). These findings offer a valuable research gap that can be further explored by other researchers in the future.

We have dived further into the approaches of Machine Learning and Deep Learning used, and we have summarized the highly used algorithms as seen in Figure 9. From the Figure, we can see a taxonomy of different categories of ML algorithms and Deep Learning. The number of studies applying a particular algorithm is denoted with a bracket For example, we have noticed that most researchers have applied the Supervised Learning type, specifically, Random Forest (39 studies), Normal Decision Tree (38 studies), Logistic Regression (35 studies), and Naïve Bayes (35 studies) as the top four (4) algorithms. As for Deep Learning, Artificial Neural Network (20 studies) and Multi-layer Perceptron Neural Network (9 studies) appeared the most among other intelligent prediction algorithms.

It is evident from this systematic review that Machine Learning is widely applied in Learning Analytics, particularly in classification techniques. In addition to the challenge of finding reliable and valid variables with significant contribution to the performance, the search for a versatile prediction algorithm with superior accuracy and validity is another challenge [139].

Most researchers opted for supervised ML algorithms due to their generality of the results and the fact that their easier to interpret such as Decision Tree and Logistic Regression.

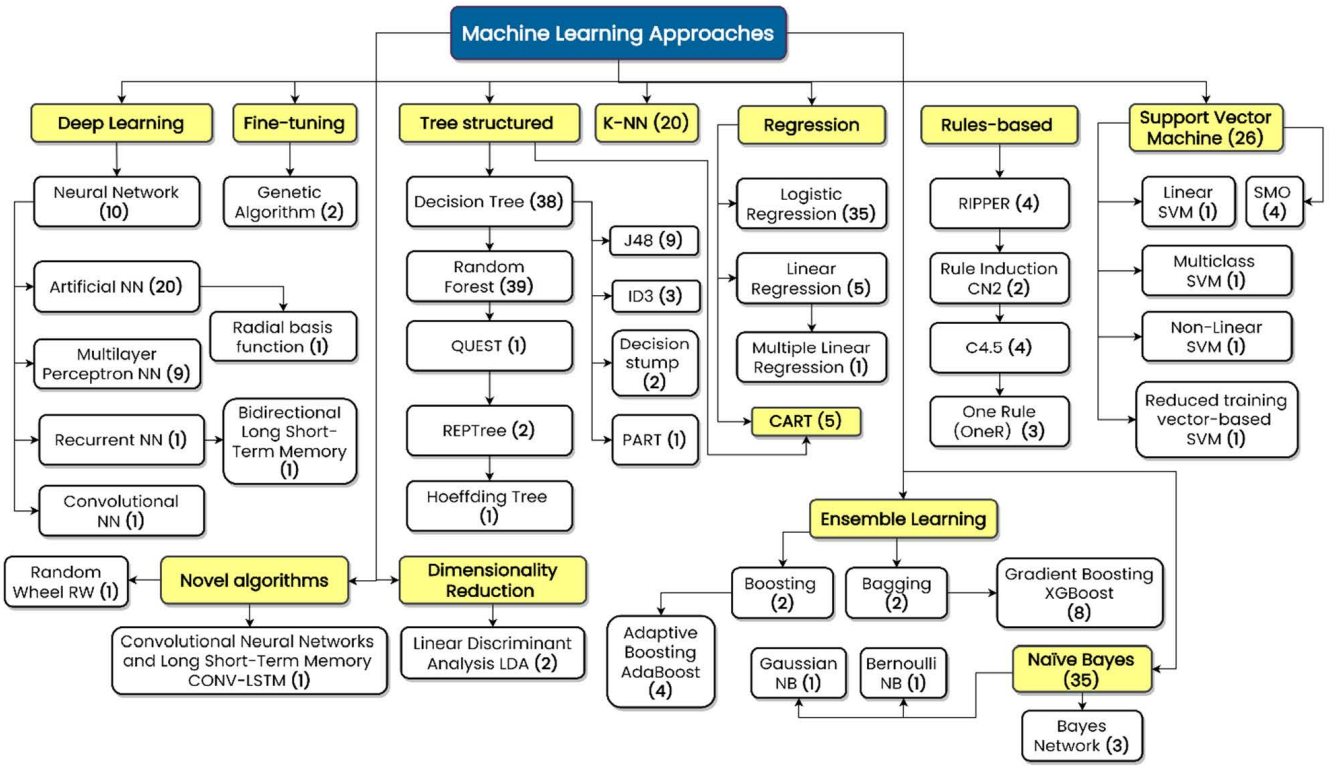


FIGURE 9. Machine learning algorithms taxonomy; presenting the number of studies applying each algorithm.

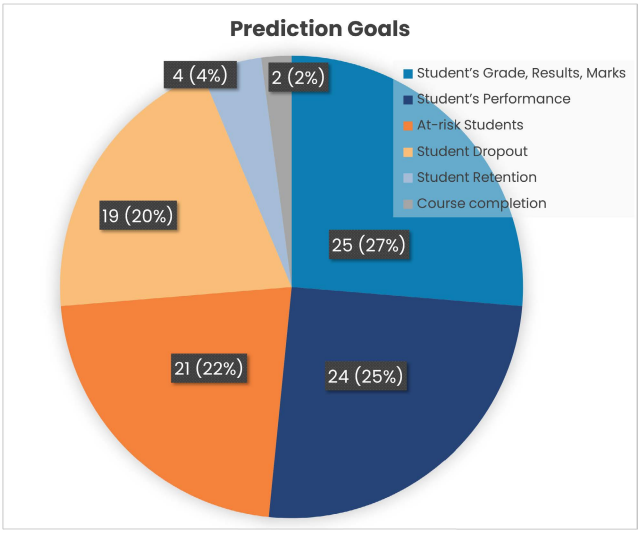


FIGURE 10. Number of Studies based on prediction goals.[1]

It is also noted that from Figure 9, researchers have applied Ensemble Learning techniques such as boosting and bagging [56], [58], [63], [73], [75], [79], [90], [104], [134]. Ensemble methods are usually used to combine several base algorithms to produce the ideal model, which in turn can increase the generalization, classification precision [137], accuracy, and robustness of the models. Deep learning techniques are widely used in LA as well. Specifically, ANN is a backpropagation algorithm, with many advantages, for example, it has great resistance to datasets that are noisy [49]. Additionally, the abilities of such algorithms to classify and find patterns for a dataset where there's no clear relation between the features and the class labels. Another type of deep learning algorithm is known as MLP, which is commonly used in situations where the structure of the problem is unclear [75].

While there are endless benefits to Supervised ML and deep learning techniques, there's still a lack of research in other intelligent areas such as clustering, semi-supervised and hybrid ML. These other approaches can help to figure out patterns that are often too subtle to be discovered by supervised and human observations. This is why it is worthwhile to explore the application of unsupervised learning, specifically clustering as it can bring benefits to identifying heterogeneous groups of students based on their behaviours and characteristics [140]. Clustering analysis can be used to group students based on similar behaviour or characteristics, Bundasak [47] implemented this method for 190 students for two technology courses and clustered them into three groups. The model analyses students' behaviour that can affect their final grades, such as spending too much time on social media and not utilizing their free time effectively, predicting this with an accuracy of 97.83-100%. It is concluded that features such as leisure (free) time and motivational behaviours significantly affect the student's learning results.

This review has also explored novel algorithms such as Random Wheel (RW) and Convolutional Neural Network and Long Short-Term Memory (CONV-LTSM). Khan *et al.* [16] focused on predicting student success and performance using the RW algorithm before the course commencement. The authors mainly focused on improving the confidence of prediction, with nearly 0.6 million records of real data used, RW achieved 66% accuracy overall, which is greater compared to the state-of-art classifiers such as DT, SVM, and RF (below 50%) and so on. The algorithm can be further enhanced using other effective predictors and making the classifier compatible with numeric attributes. Several attributes indicating the performance of students are considered such as student's Cumulative Grade Point Average (CGPA) and historical grades, and past performance in similar courses. Additionally, the teaching evaluation is also included which indicates this is a student and teacher-centred approach. However, higher accuracy can be achieved by exploring the behavioural attributes of the students.

Mubarak *et al.* [27] focused on addressing the student dropout issues by proposing a CONV-LSTM algorithm. This is based on the deep learning technique using data logs obtained from Massive Open Online Courses (MOOCs) for raw clickstream. Usually, the dropout rates in such courses seem to be high with approximately 25% course completion for devoted students. The model achieved an area under the curve (AUC) rate of 85% and 86% for Dataset 1 and Dataset 2, respectively. Certain features were considered such as online activities, navigation, viewing videos/pages, and so on. The difference in this approach compared to machine learning-based approaches is that the authors here applied automatic feature extraction instead of manual feature engineering done by specialists, which can cause biased results. This research also addresses the class imbalance problem using the cost loss function.

Furthermore, from Figure 8, we noticed that Analysis and Statistical approaches are also widely used in LA. Descriptive (percentages, frequencies, means, and standard deviations) and inferential statistics (correlation) are used for the analysis of student performance data such as Analysis of Variance (ANOVA), Paired T-Test, Automatic Linear Modelling (ALM), Unpaired Two Sample Wilcoxon Test, and Pearson Coefficient Correlation.

In Figure 10, we explore the different prediction goals obtained by the reviewed studies in Learning Analytics. We identified six (6) main prediction goals. Firstly, mostly the predictive models were developed to predict the assessment outcome of students (25 studies), such as the final Grade, assessment results, or final marks. The second top prediction goal is student performance (24 studies), where the model predicts whether the student "passes" or "fails". Thirdly, at-risk students (21 studies), where the student categorized as either "at-risk", "not at-risk", and so on. Fourthly, student dropout (19 studies). Other prediction goals include student

[1]Other prediction goals include Learning style, Successful submission of student's tasks, Low-Engagement students, Online learning outcomes, and Student status with one study for each goal.

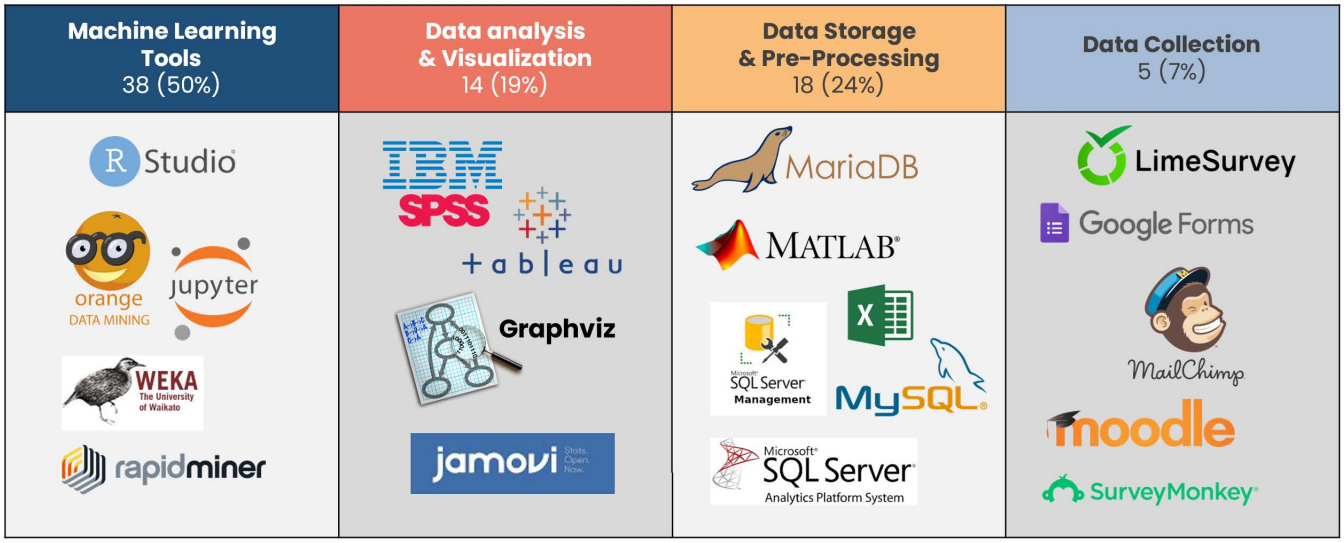


FIGURE 11. Available tools for learning analytics; presenting the number of studies for each category.

retention (4 studies) and course competition (2 studies). The remaining prediction goals include Learning style (1 study) Successful submission of student's tasks (1 study), Low-Engagement students (1 study), Online learning outcomes (1 study), and Student status (1 study).

We have also dived into the type of the final predictor outcome and its type as can be seen in Table 6. The Table also provides the corresponding studies for each predictor and the type. Our findings reveal that most studies have predicted categorical types instead of numeric.

Figure 11 depicts the tools used by researchers in the Learning Analytics field, extracted based on the findings from this survey. It is worth noting here that the depicted numbers represent only the articles that mentioned tools regarding the data collection, management, analysis, visualisation, and model development. Hence, 25 out of 100 articles have not mentioned the implemented tools. Nearly 50% of studies have used Machine Learning tools, 19% have used data Analysis and visualization software, 24% have used data storage and pre-processing, and finally, 7% of studies have used data collection tools. As previously mentioned in section B, figure 4, most researchers often opt for extracting records available from the university. In another way, records that demonstrate students' online behaviour can be extracted using different Moodle Plugins from the directory (such as Attendance Register, Configurable Reports, and Analytics Graphs) [83]. Additionally, email can be another source for data collection which can be done using tracking tools such as Mailchimp e-mail tracking platforms [83]. Furthermore, researchers further extended the data collection methods through surveys and questionnaires, usually, gathered with the help of survey tools such as LimeSurvey [52], Google Forms [52], [115] and SurveyMonkey [86].

In terms of languages used, our survey found that Python is among the top programming language used with 55% of researchers making use of this language. The second-highest programming language is R (25%). Other languages include JAVA (15%) and SQL (5%). For tools used to prepare data and implement the predictive model, we noticed that most researchers used Waikato Environment for Knowledge Analysis (WEKA) ([16], [20], [98], [101], [104], [108]–[111], [113], [121], [129], [47], [50], [52], [54], [71], [76], [93], [94]) for ML approaches as it's the top tool with 27% of studies utilizing this. The second highest is IBM SPSS for statistical approaches with 13% ([13], [66], [70], [82]–[84], [86], [95], [120], [136]). Other most-used tools include Microsoft Excel (17%) ([20], [52], [117], [123], [135], [71], [82]–[84], [86], [89], [112], [115]), RapidMiner Studio (9%) ([13], [49], [70], [86], [91], [107], [131]), Jupyter Notebook (7%) ([115], [123], [124], [127], [132]), Orange and R Studio (4%) ([62], [87], [116], [125], [128], [133] respectively), and Tableau (3%) ([77], [127]).

Python is a well-known programming language with increased popularity compared to R. It is mainly used for general programming, especially Machine learning in large-scale web applications, while R is used for statistical computing and graphics. R usually requires more lines of code [141] compared to Python. Other tools used by researchers in Learning Analytics are summarized in Figure 11.

D. QUALITY ASSESSMENT RESULTS

This systematic review evaluates the quality of the surveyed articles based on eight (8) critical criteria. Four (4) guidelines were adopted from [40] review with the addition of four (4) new criteria to conclude a valuable research gap and elaborative answers to our formulated research questions. The added criteria are authors address and resolve ethical issues (C02), models and algorithms address data imbalance problem (C03), authors apply Feature Engineering (C04), and the sample or dataset size includes more than 1000 instances (C08).

TABLE 6. Learning prediction outcome & type.

#	Learning Prediction Outcome & Type		References	No. of studies
1	Categorical	Dropout [Dropout, Did Not Dropout]	[10], [13], [120], [123], [125], [129], [132], [134], [23], [51], [56], [92], [93], [109], [110], [116]	17
2		Performance classes [Pass, Fail]	[16], [20], [82], [85], [87], [101], [102], [112], [126], [52], [58]–[60], [65], [71], [76], [77]	17
3		Student Status [Good, Weak]	[49]	1
4		Retention [Retained, Not Retained]	[119], [127]	2
5		Results [A+ to F]	[99][75][73][113][55]	5
6		Risk Status [at-risk, not at-risk]	[61], [69], [83], [106]	4
7		Student Relative Classes [group 1, group 2…]	[47], [88], [96]	3
8		GPA Prediction [Good, average, poor]	[55], [98], [122], [131]	4
9		Student level [Medium, High, Low]	[44], [70], [91], [94]	4
10	Numeric	Grade Prediction [CGPA, marks]	[55], [56], [62], [104], [115]	5

TABLE 7. Available datasets for learning analytics.

#	Available Dataset & References	Online Source	No. of Studies
1	Student Performance Data Set ([44], [55], [91], [104], [114], [126], [128])	https://archive.ics.uci.edu/ml/datasets/student+performance	7
2	Student-Drop-India2016 [120]	https://www.kaggle.com/imrandude/studentdropindia2016	1
3	S1 File. Data from mathematics and computer science degrees. (grades) [56]	CSV File - https://journals.plos.org/plosone/article/file?type=supplementary&id=info:doi/10.1371/journal.pone.0171207.s008	1
4	S2 File. Data from law degree (grades) [56]	CSV File - https://journals.plos.org/plosone/article/file?type=supplementary&id=info:doi/10.1371/journal.pone.0171207.s009	
5	Open University Learning Analytics Dataset (OULAD) ([58], [59], [65], [68], [70], [72])	https://analyse.kmi.open.ac.uk/open_dataset	6
6	Early-Warning-System [135]	https://github.com/FelipeNb/Early-Warning System/tree/master/modules/Prediction	1
7	Students-Dropout-Prediction [78]	https://github.com/JKabathova/Students-Dropout-Prediction	1
8	Students' Academic Performance Dataset (xAPI-Edu-Data) ([61], [126], [128])	https://www.kaggle.com/aljarah/xAPI-Edu-Data	3

Table 8 highlights the results of the quality evaluation. It presents the number of studies that satisfied the criteria in each learning environment, traditional, online, and blended. It is observed that few researches conducted with the consideration of ethical methods ([20], [23], [80], [83], [84], [86], [92], [95], [97], [107], [120], [129], [45], [50], [51], [60], [62], [65], [70], [78]), precisely a total of twenty (20) studies out of hundred (100) address and resolve ethical issues. Secondly, addressing data imbalance issues is crucial in nearly all machine learning algorithms and can potentially increase the accuracy of the model if resolved ([16], [27], [81], [92], [120], [121], [124], [127], [129], [134], [51], [56], [57], [64], [68], [69], [78], [79]). However, we can see from the Table that only twenty-one (21) studies addressed this. Lastly, only fourteen (14) of the studies acknowledged that the obtained results were verified using a second dataset, on unseen data other than the training and testing sets ([16], [27], [121], [129], [130], [135], [46], [50], [61], [63], [104], [106], [117], [120]), which may imply the generalizability of results.

We also observed from the Table that a satisfactory number of studies (83) applied Feature Engineering (FE) techniques. This is an essential part of data pre-processing to further enhance the prediction accuracy, as proven by surveyed studies [127], [128]. This process can construct new features from a dataset based on the researcher's knowledge of the domain and different data manipulation techniques. However, manual FE can cause bias if unnecessary or insignificant features are obtained.

V. DISCUSSION AND CRITICAL REVIEW

In this section, further discussion of the reviewed articles will be provided. The conducted survey revealed that the use of learning analytics approaches to predict student's performance and enhance the learning methods in education has been on the rise since the year of 2017 till today in both developed and developing countries. To further analyze the included studies, the findings are summarized to answer the research questions presented in section III.

TABLE 8. Results of the quality assessment of surveyed studies; presenting the number of studies that satisfied the assessment criteria.

Assessment Criteria	*Number of Studies Fulfilling the Criteria*		
	TLE	OLE	BLE
Data collection methods are clearly stated **(C01)**	46	26	22
Authors address and resolve ethical issues **(C02)**	**10**	**5**	**5**
Models and algorithms address the data imbalance problem **(C03)**	**9**	**7**	**5**
Authors apply Feature Engineering **(C04)**	41	22	20
The predictor variables considered are clearly identified and described **(C05)**	46	24	15
Authors make use of separate (split dataset) training and testing datasets **(C06)**	35	20	16
The predictive model is tested on an unseen dataset or verified with a second dataset **(C07)**	**8**	**4**	**2**
The sample or dataset size includes more than 1000 instances **(C08)**	**14**	**9**	**3**

* *TLE, OLE and BLE stand for Traditional Learning Environment, Online Learning Environment, and Blended Learning Environment respectively.*

Here, we will provide answers to **RQ1** to determine the factors that can help determine high-risk students and the factors that fail to predict those students. It is observed that in prediction, the number of variables is not important, however, the quality of the variables can highly contribute to the performance and accuracy of the prediction model [10]. Therefore, the variables must be carefully selected to address a specific prediction problem such as the performance or success of the students.

- **Academic factors**: Our findings reveal that most studies used traditional and common factors to predict student performance. Our findings also reveal that academic results are appropriate data for dropout prediction. For example, the ''grade'' attribute appeared a hundred and six (106) times in the reviewed articles. While this attribute is highly important to predict the performance of the student, it is still considered an insufficient indicator of why the student is performing in this way or that level. Therefore, it only answers the question of how the student is performing in each assessment; not why he/she fails or succeeds. Furthermore, while the ''attendance'' attribute can help indicate the determination of successful students, it can fail to accurately identify high-risk students as attendance might be influenced by several factors which can be beyond the student's control, for example, health issues, family commitments or unexpected work events, etc.
- **Time-related factors:**features such as ''time spent in an active session'' or ''time spent editing a wiki'' have been proven to have a low correlation or insignificant effect on the student performance as the student might log in and leave a session open for a long time without interacting with the resources and activities in the system.
- **Demographic/socio-demographic factors:** variables such as ''age'', ''address'', ''phone number'', and ''gender'' are often omitted from the training set as they are known to be static variables. Researchers have proven that such variables are less statistically significant in predictive models.

A summary of the main categories and sub-categories variables identified in this review is presented in Appendix A. From the Figure, we can see the number of times the feature has occurred in the reviewed studies. Additionally, the insignificant variables are denoted by an asterisk (*) to highlight the features that are not indicative of student

TABLE 9. Summary of the student data sources; denoting significant and insignificant features.

Data Sources in Learning Analytics				
Main-category	Sub-categories			
Personal	Hobbies (1)	Academic interest (3)	Time spent on social media, and watching movies (4)	Free time (10)
	Social skills/talents (3)	Management skills (2)	Personality traits (1)	Internet access (8)
Academic	**Attendance (27) ***	Class Performance/failures (13)	extra-curricular activities (11)	Intak year/cohort (11)
	Year/sem repetition (1)	Plagiarism Count (1)	Exam attempts (2)	
Economic & Financial	Fees payment (2)	Institutional Financial Aid (3)	Loan details (1)	
Marks & grades	**GPA/CGPA (25)**	Subject/course grades (19)	**Final grade/marks (30)**	Sem marks/results (3)
	Continuous assessment marks (17)	Exam marks/grades (12)		
Demographics	**Gender (47) ***	**D.O.B/age (24) ***	Address (15)	Marital status (13)
	Name (3)	Race/ethnicity (5)	Country of origin (14)	
Student's Health	Current health status (8)	Alcohol consumption (6)	Health insurance (1)	medical condition (4)
	Disease (1)			
Family-related	Family income (13)	Family size (6) *	Parent's cohabitation status (7)	Parent's education (14) *
	Student's guardian (11)	Family educational support (7)	parent's involvement (4)	Family trouble (1)
	Family Assets (1)	Parent's job (8)	Family Expenditure (1)	
Employment	Career goals (2)	Internship Details (1)	Employment status (4)	
Behavioural	Class participation (1)	Effect of friend's circle (1)	Interaction with faculties (1)	Attention in academic studies (2)
	Motivation factors (3)	School violations (1)	Help-seeking (1)	Raised hands (7)
Studying & Learning Style	Study time (10)	Studying style/skills (2)	Use of extra learning materials (2)	Tutoring (1)
	Procrastination (1)	Reading habits (1)	Learning styles (1)	
LMS Log Data	Total number of logins (10)	**Resources/course access (22)**	Task/assignment submission (15)	Event date/time/type (6)
	Clickstream (avg, sum of clicks) (6)	Video interaction (rewind, play, pause, like, dislike) (8)	E-book actions (3)	IP Address (1)
	Interactions with peers/instructor (2)	Instructor's actions (posting/grading etc) (1)	Time spent (10) *	Number of downloads (1)
Pre-University Data	School grades (14)	School name/type (6)	Graduation year (1)	
Teaching evaluation	Tutoring support (1)	Student satisfaction (1)	Parent satisfaction (7)	Student's expectation (1)
	Student perception (1)			
Lecturer/teacher/ supervisor profile	Characteristics (1)	Qualifications (1)	historical teaching evaluation score (1)	

Legend: >= 20 studies | 10 - 19 studies | 9 - 5 studies | < 5 studies

() Denotes number of feature appearance in studies

* Denotes features proven to be insignificant in Learning Analytics

TABLE 10. Summary of the related work in Traditional Learning Environment, sorted by dataset size.

#	Ref.	Year	Prediction Goal	Dataset Size/ No. of Variables	Input Data	Algorithm(s)	Approach	Finding(s)	Limitation(s)
1	Muncie [127]	2020	student retention	179,517 / 15	Administrative Student data	LR, NB, and DT.	Supervised ML	• Predicts at-risk students with 80% probability and above. • Feature engineering improves the accuracy of results.	Oversampling increases data noise with overfitting risks.
2	Lee & Chung [51]	2019	Student Dropout	165,715 / N.A.	Behavior, Family, Maladjustment & health	RF & Boosted DT	Supervised ML	• Boosted DT resulted in the best performance.	Overfitting due to oversampling
3	Mduma [120]	2020	Student Dropout	110356 / 20; 19101 / 15; 145 / 12.	Demographics, Student's health, Reading habits, Parent's involvement	Linear Regression, Multi-Layer Perceptron (MLP), RF & KNN	Deep Learning	• Builds a web-based prototype of a prediction system to record student dropout • Student gender has the strongest contribution (**20%**) to drop out prediction, higher dropout rate for girls. • Logistic Regression and Multi-Layer Perceptron are the best classifiers.	Fails to study the behavior and social factors of students
4	Hussain & Khan [45]	2021	student's mark & grade	90,000 / 126	Obtained marks and Grades	Genetic Algorithm (GA), DT (Regression & classification)	Supervised ML	• Applies auto attribute selection using GA. • DT – classifier resulted in **94.39%** accuracy; DT – regression attained RMSE of **5.34**.	Doesn't resolve data imbalance issues
5	Palacios et al. [129]	2021	At-risk students	6656 / 165	Demographic, geographic, school & university performance, financial	DT, k-NN, LR, NB, RF, SVM	Supervised ML	• Random forest (accuracy: **81.82%**) performs the best compared to other techniques. • Reveals that balancing the classes in models improves accuracy.	The large difference in the number of instances used to train the learning models
6	Gray [50]	2019	student retention	4970 / 32	Student engagement and academic performance	Random Forest	Supervised ML	• Achieves an accuracy of more than **97%.** • Attendance and Academic year contributed the least to the prediction of failure students	Doesn't address data imbalance problems.
7	Rovira et al. [56]	2017	Student Dropout & Grade Prediction	4,434 / N.A.	Student's grades	SVM, AdaBoost, RF, NB, LR	Supervised ML	• Easily adaptable and generalizable since the model considers only "grades". • The classification system achieves an F1 score of **82%**.	Does not explore student's behavior
8	Marwaha & Singla [46]	2020	At-risk students	2560 / 30	Academic demographic, social, & behavioral factors, semester performance	DT, ID3, CHAID, NBC RI, RF & EN	Supervised ML	• Ensemble methods give the best accuracy (**86.67%**)	Doesn't resolve data imbalance issues
9	Jeslet et al. [112]	2021	student's grade	1460 / 15	Final year student results	KNN, SVM, LR	Supervised ML	• SVM and Logistic Regression resulted in an accuracy of **99.72%** compared to KNN.	Doesn't resolve data imbalance issues
10	Yousafzai et al. [55]	2021	student's grade	1044 / 33	Demographics, Student's health, Reading habits, Parent's involvement	Bidirectional Long Short-Term Memory (BiLSTM)	Deep Learning	• The model results in accuracy between **88-90%**	Doesn't consider social & cultural attributes
11	Lozano et al. [10]	2020	Student Dropout	935 / N.A.	Sociodemographic & Academic Data	Classification & Regression Trees (CART) and Quick, Unbiased, Efficient, Statistical Tree (QUEST) algorithms	Supervised ML	• A longer-term and continuous retention strategy (accuracy rate: first model: **60.9%.** Second: **70.9%.** Third: **76%.** • The First year is the appropriate time to identify high-risk students	The approach may not be holistic (student-centered only)
12	Nouri et al. [95]	2019	course completion	900 / N.A.	Student & supervisor performance, experience & research output	NB, LR, GBT, NN, Deep Learning, DT, & RF	Analysis and Statistical approaches & DL	• Logistic Regression resulted in the highest accuracy (**73.5%**) • There's a strong correlation between "supervisors' non-completion of thesis history" with students' incomplete thesis.	Does not consider behavioral and cognitive factors
13	Feng [125]	2019	Grades, results, marks	800 / N.A.	Demographics, Academic Performance	Analysis of Variance (ANOVA), Decision Tree (DT), ANN	Analysis and Statistical approaches and DL	• Neural networks outperform DT classifier. • Model results in **92.8%** accuracy rate	Doesn't address data imbalance problems.
14	Daud et al. [109]	2017	Student Dropout	776 / 23	Family expenditure, Income, Personal & Family Assets	SVM, C4.5, CART, Bayes Network & NB	Supervised ML	• Features such as family expenditure and personal information significantly impact the student's performance	It may be biased towards the max class values
15	Jimenez et al. [130]	2020	Student Dropout	753 / N.A.	socio-demographic, program data & student's performance	RF	Supervised ML	• Proves that sociodemographic variables do not significantly affect the prediction of student dropout. • Results in prediction accuracy of **91-99%**	May result in slow prediction
16	S. Priya et al. [44]	2021	student's grade	(UCI) 649 / 33	Student Marks, Social, Demographic Data and School-Related Features	ANN, SVM, LR	Deep Learning	• Logistic Regression resulted in the highest accuracy of **67-69%**.	Focuses on academic attributes only
17	Pallathadka et al. [114]	2021	student performance	(UCI) 649 / 33	Academic results	NB, ID3, C4.5, and SVM	Supervised ML	• Students' talents and interests might be linked to their performance. • SVM is the most accurate (**85%**) technique for classifying student performance.	Doesn't resolve data imbalance issues

TABLE 10. *(Continued.)* **Summary of the related work in traditional learning environment, sorted by dataset size.**

18	Arhin & Laryea [48]	2020	student retention	625 / 11	Tutor's characteristics & student's perception	LR	Analysis & Statistical approaches	• A student with a positive perception of tutoring support has an **83%** chance of staying in the programme.	Not generalizable
19	Hashim et al. [20]	2020	student's final grade	499 / 8	Demographics, employment, Academic Achievement (grades)	DT, NB, LR, KNN, MLP, SMO, NN, PART, JRip and OneR	Deep Learning	• Logistic regression classifier is the most accurate (**95.4%**) in predicting the exact final grades of students. • Large dataset increases the accuracy.	Dataset size affected the accuracy
								• Less predicted values increase the accuracy (6 values = 68%, 2 values = 80%)	
20	Bujang et al. [113]	2021	student's grade	489 / 11	Demographics & Grades	J48, LR, RF & SVM	Supervised ML	• J48 (**99.8%**) is suitable for the prediction of student's final grade	Limited dataset & attributes
21	Aziz [91]	2020	student performance	450 / 17	Demographics, Student's health, Reading habits, Parent's involvement	Deep Learning, RF & Gradient Boosted Trees	Deep Learning	• Deep Learning is the most suitable algorithm predicting performance.	Oversampling may increase data noise.
22	Ahammad [75]	2021	student's grade	400 / 1	Marks	NB, KNN, SVM, XGBoost & MLP	Deep Learning	• MLP achieved the highest accuracy **86.25%**.	Limited dataset, only marks attributes are used.
23	Athani et al. [108]	2017	student's grade	(UCI) 395 / 34	Academic Performance and Social Behavior	Multiclass Support Vector Machine (SVM) & Neural Networks (NN)	Supervised ML	• Multiclass SVM performed better with an accuracy of **89%.**	Doesn't address data imbalance problems.
24	Veerasamy et al. [117]	2020	student's final grade	289 / N.A.	Formative assessment tasks (homework & demo exercise) & final exam results	Classification tree analysis (manual) & RF	Supervised ML	• Model results in **60%** prediction accuracy overall. • Predict failure students as early as Week 3. • Formative assessment (homework) partially affects student success.	Not generalizable and limited sample size
25	Iqbal et al. [122]	2020	student's grade	180 / 13	Gender, Family Related, Educational and Personal, and Student's Grades.	Linear Regression, DT Regression, Gaussian NB, and DT Classifiers.	Supervised ML	• Decision Tree Classifier results in better accuracy of 95-100%.	Limited to one course; does not address class imbalance problems
26	Altabrawee et al. [49]	2019	student status	161 / 20	Personal And Lifestyle, Studying Style, Family Related, Educational Environment Satisfaction, and Student's Grades.	ANN, NB, DT, and LR.	Deep Learning	• ANN performs better (**80%**) with high classification accuracy (**77.04%**). • Course grade related to computer, Accommodation, interest in studying the course, educational environment Satisfaction, and the residency contribute significantly to classifying student's status.	Limited to one course; results are not generalizable.
27	Gil et al. [93]	2020	Student Dropout	150 / 12	Students' Grades, Demographic, and Academic Performances	NB & Decision Tree (J48)	Supervised ML	• J48 performs better than NB with an accuracy of **94%**.	Limited Sample of Students
28	Dabhade et al. [115]	2021	student's final grade	85 / 10	Personal, education, behavioral, extra-curricular, GPA & admission	SVM, Linear Regression	Supervised ML	• Multiple LR shows the best fit with an R^2 score of **83.27%**.	Limited dataset size
29	Yaacob et al. [116]	2020	Student Dropout	64 / 27	Students' Grades & Demographics	DT, LR, RF, KNN & NN	Deep Learning	• Logistic Regression model is best to predict the dropout students.	Limited Dataset used. Model validation is
									lacking. Not generalizable.
30	Khan et al. [94]	2019	student's grade	50 / 12	Demographic, Academic & Grades	NB, SMO, MLP, IBK, Dec. Table, JRIP, OneR, ZeroR, J48, RF, Random tree	Deep Learning	• DT algorithms (J48) resulted in the highest accuracy with a rate of **88%**. • CGPA, test marks, and Attendance are strong predictors	Overfitting due to oversampling; limited sample size
31	Maheswari et al. [85]	2021	student performance	N.A.	Demographic, financial, parent's involvement & results	KNN	Unsupervised ML	• Result, attendance, and class performance contribute significantly to students' performance (accuracy **63%**). • Overall accuracy result is **66.5%**	Uses synthetic data
32	Kemper et al. [92]	2020	Student Dropout	N.A.	Demographics and Academic Achievement (grades)	LR and DT	Supervised ML	• Model and data preparation are less complicated in DT than LR. • Achieves an accuracy in classification of more than **83%** after semester 1.	Results may not be generalizable due to specific course
33	Chauhan et al. [118]	2019	student's grade	N.A.	Theory Marks, Term Test Marks, Viva, and Practical Marks	kNN, DT, SVM, RF, Linear Regression	Supervised ML	• Multiple Linear Regression performs better.	Only focuses on grade's attributes

* *(N/A) denotes unavailable information; Based on the author's knowledge.*

performance and the understanding of student behaviours. It is found that using more features may reduce the bias in training but possibly increase the variance that causes an algorithm to capture the random noise in the training data, which cannot be generalized to other samples. **RQ2** and **RQ3** have been addressed in the preceding section. A comparative summary of the reviewed studies is included in Appendix B, due to the constraint of pages; the authors selected some articles out of the 100 studies for each learning environment. The decision to exclude some articles was made based on the

TABLE 11. Summary of the related work in online learning environment, sorted by dataset size.

#	Ref.	Year	Prediction Goal	Dataset Size/ No. of Variables	Input Data	Algorithm(s)	Approach	Finding(s)	Limitation(s)
1	Wang et al. [99]	2018	student performance	2607207 / 158	Academic, demographics, and online behavior	Decision Tree	Supervised ML	• Results in a predictive probability 70.25% to 95.1%. • Develops an intervention/ feedback system	Doesn't address data imbalance problems
2	Chen et al. [90]	2020	At-risk students	64,754 / N.A.	Online behaviors	RF, J48, LR & Bagging	Supervised ML	• Results in the precision of **90.4 to 93.6%.** • Able to predict at-risk students at different phases of a semester.	Cognitive features are not considered
3	Adnan et al. [58]	2021	at-risk student	32593 / 31	Demographics, VLE interaction & assessments	RF, SVM, K-NN, ET, AdaBoost, Gradient boosting, and ANN	Deep Learning	• Students who completed and passed their course are more involved with VLE activities.	Doesn't address data imbalance problems.
4	Rayasam [123]	2020	Student Dropout	15,000 / 9	Gender, Ethnicity, Major, Leave of Absence	Logistic Regression Algorithm	Supervised ML	• Address mental health problems among undergraduates • Logistic Regression can predict dropout with 69.3% certainty	Doesn't address data imbalance problems.
5	Bravo-Agapito et al. [88]	2021	Early prediction of	802 / 25	LMS Log Data, Demographics. &	Multiple Linear Regressions, Cluster Analysis	Analysis and Statistical approaches	• Student performance is inversely proportional to age.	Results are not generalizable
			student performance		Academic Achievement.			• Number of interactions with Moodle is closely related to student performance. • **"Time spent" on a task is not an important predictor.**	
6	Hasan et al. [62]	2020	Student's results	772 / 19	Academic, LMS (online activity), interaction (video)	Classification Tree, RF, KNN, SVM, LR, NB, NN & CN2 Rule Induction	Deep Learning	• Random Forest achieved a prediction accuracy of **88.3%**	Doesn't address data imbalance problems.
7	Queiroga et al. [134]	2020	Student Dropout	752 / N.A.	VLE interactions logs	DT, RF, MLP, LR & AdaBoost (ADA).	Analysis and Statistical approaches & Deep Learning	• Results in a generalized model solely based on VLE interactions. • Predicts at-risk students before the end of the first semester. • Use of GA algorithm to optimize hyperparameters can improve accuracy.	Not generalizable
8	Soffer et al. [66]	2019	course completion and student grades	646 / 13	Academic and Online activities	ANOVA, Logistic Regression and Hierarchical linear regression	Analysis and Statistical approaches	• Students who completed the course had higher online engagement than non-completers; depending on the course subject.	Results may be biased as it does not study student and instructors' characteristics.
9	S. Vairachilai & Vamshidharreddy [61]	2020	student performance	480 / 16	Demographic, Academic & Behavioral.	DT, SVM, and NB.	Supervised ML	• Naïve Bayes resulted in **77%** accuracy compared to DT and SVM.	Doesn't address data imbalance problems
10	Akour et al. [67]	2019	student performance	480 / 16	Demographic, Academic & Behavioral	Convolutional neural network (CNN)	Deep Learning	• Results in accuracy of **95.5%** (no. of epoch = 200)	Limited dataset size
11	Hussain et al. [70]	2018	Low-Engagement Students	384 / N.A.	Demographic, Grades & learning behavior in VLE	J48, DT, JRIP, CART, GBT and NB	Analysis and Statistical approaches & Supervised ML	• Final Grades and the number of clicks in VLE have a significant and positive correlation. • Proposes an instructor dashboard with two main components: a prediction model and a data visualizer. • DT (**85.91%**), J48 (**88.52%**), JRIP (**83.27%**), and GBT (**86.43%**) are the best performing algorithms	Doesn't address data imbalance problems
12	Umer et al. [74]	2017	student performance	167 / 10	Assessment & Final Grades, Submission Time, Video Lecture Interaction Log, Demographic,	Naive Bayes, RF, LR, KNN & paired t-test	Analysis and Statistical approaches & Supervised ML	• NB (**89%**) classifier outperforms other techniques. • Time Spent Weekly and Video Watch Activities are the most significant to predict student performance.	Limitations of the dataset exist limited size, missing values, and imbalanced classes.
					Time Spent Weekly				
13	Rao & Kumar [57]	2021	student performance	108 / N.A.	Online Activities – LMS logs (viewing, submission, attendance & quiz)	Recurrent Neural Network (RNN)	Deep Learning	• The proposed model achieved an accuracy of 84.3%	Limited dataset size
14	Sultan et al. [23]	2021	Student Dropout	N.A.	Questionnaires Data	Random Forest, Decision Tree, Logistics Regression, SVM, Naïve Bayes	Supervised Machine Learning	• Random Forest is the best classifier with an accuracy rate of **97.92%**	Doesn't resolve class imbalance problems
15	Waheed et al. [59]	2020	at-risk student	N.A. / 59	Activities and demographics in VLE (OULAD)	Artificial Neural Network	Deep Learning	• ANN model outperforms SVM and Logistic Regression with an accuracy of **84%–93%.**	Results may be biased due to imbalance issues

TABLE 11. *(Continued.)* **Summary of the related work in online learning environment, sorted by dataset size.**

16	Huang et al. [63]	2020	At-Risk Students	N.A.	Online Learning behavior & e-book reading logs	GaNB, SVC, linear-SVC, LR, DT, RF, NN, and XGBoost	Deep Learning	• Both DT (**92.8%**) & NN (**97.9%**) work well to predict student's performance	Imbalanced data for learning actions, thus affecting model performance.
17	Chen & Shiri [64]	2020	At-Risk Students	N.A.	Moodle logs: Raw log data and gradebook	Gradient boosting machine (GBM), RF, KNN, DT, SVM, NN, NB, LR	Deep Learning	• Top features predicting the performance in courses are quiz and assignment results. • Results obtained show that GBM, RF, and the ensemble models performed the best.	Not generalizable may cause overfitting & increased noise
18	Bañeres et al. [60]	2020	at-risk student	N.A.	Student's Grades	DT, KNN, NB Bayes, SVM, Unpaired two-sample Wilcoxon test	Supervised ML & Analysis and Statistical	• Gradual At-Risk (GAR) model develops a threshold to identify the level of risk and provide feedback accordingly.	Doesn't address data imbalance problems; doesn't consider students' behavior
19	Neves et al. [135]	2019	Student Dropout	N.A.	LMS Moodle (Content publishing, creating ca comments, viewing the course, file submission)	Decision Tree, KNN, SVM, Random Forest, Logistic Regression, and Multi-Layer Perceptron	Deep Learning	• The proposed ensemble model outperforms other classifiers (**93%**)	Doesn't consider behavioral factors of students
20	Helal et al. [71]	2018	Student's performance	N.A.	Socio-demographic, academic & activity data from Moodle	NB, SMO, J48, and JRip,	Supervised ML	• Considering combined features rather than solo features help in predicting unsuccessful students	No diversity in students' data

** (N/A) denotes unavailable information; Based on the author's knowledge.*

''Limitation(s)'' column, whereby if there are articles with similar findings for the limitations, we chose the most recent to be represented in the appendix tables until the required number of pages is met.

A. SUGGESTIONS FOR FUTURE RESEARCH

Based on this systematic review of recent studies on the topic of Learning Analytics, we were able to point out several limitations and underexplored methods that could help build better predictive models to understand students and their behaviours in education, prevailing in the existing learning outcomes prediction models. Below are the limitations of the existing studies that are being explored by future researchers:

1) ***Results are not generalizable.*** Many aspects can reduce the generalizability of the predictive models presented such as focusing on a single course in a university or school. Each course differs from one another in the way the assessments are presented and marked. Therefore, building a comprehensive predictive model is required by considering features that can be shared by students across different courses in an institution. Another approach could be to use data from courses that share the same teaching and learning methods, assignment marking system and course structure. Lastly, future researchers should also consider using low-cost variables that do not always require excessive effort in data collection.
2) ***Data imbalance and overfitting problems.*** This is one of the most common limitations in the current studies. It is a fact that most machine learning algorithms work best when the data is distributed, so when the majority class is lower than the minor class and vice-versa, it will result in a biased model. Although some researchers tried to resolve this issue using sampling techniques such as Synthetic Minority Oversampling Technique (SMOTE), there are still pitfalls to these methods such causes increasing the noise with overfitting risks; and in the case of under sampling, there will be a loss of valuable information. While these methods are applied at the data level, other methods can be explored such as using real data only instead of synthetic data, or using a customized Cost-Sensitive Loss Function, or ensemble methods which is at the algorithm level and can result in more accuracy of the predictions model and fair conclusions of the derived patterns.
3) ***Only a student-centred approach is considered***, whereas educators' details can also be considered for the prediction of student performance and understanding why they fail to achieve their goals. Factors such as teachers' qualifications and the number of subjects may result in promising results for the predictive models.
4) ***Limited sample size***, as discussed in previous sections, ML requires a huge amount of data to accurately train the prediction model, most studies here used a minimal sample size of below 1000 instances, which is insufficient to draw out accurate conclusions and interpretation of the predictive results. Model training can be improved based on the dataset/sample size, the more data, the more accuracy and therefore, clear patterns can be concluded from the prediction outcome.
5) ***Traditional student performance factors*** such as grades, attendance, economic, and demographic are considered ignoring other vital factors such as personal, cognitive, and behavioural factors. Other features that can be explored are teacher-related such as

TABLE 12. Summary of the related work in blended learning environment, sorted by dataset size.

#	Ref.	Year	Prediction Goal	Dataset Size/ No. of Variables	Input Data	Algorithm(s)	Approach	Finding(s)	Limitation(s)
1	Ho & Shim [102]	2018	at-risk student	2500 / N.A.	Student Profile Data, Student Activity Data (Online), Learning Context Data (Module Info)	LR, NB, KNN & RF	Supervised ML	• Factors such as "assessment results & online interaction" are good predictors for the outcome of the module.	May not be able to predict students with "fail" outcome
2	Jokhan et al. [80]	2019	Early prediction of student performance	1523 / N.A.	Moodle logs: online activities (login frequency of students, user interaction patterns, completion rate, and the coursework status of students)	Regression Model & ANOVA	Analysis and Statistical approaches	• Predicts student performance based on their login activity and completion rate. • Results in prediction accuracy of **60.8%**	Does not consider non-cognitive attributes such as financial data and family-related matters
3	Albassam [121]	2019	at-risk student	1,476 / N.A.	VLE: Viewing Module, Resources, Creating & Viewing Posts, VLE access patterns	Logistic Regression	Supervised ML	• Builds a multi-course early warning Adaptive Grey Zone Model (AGZEM). • Results in 87% prediction accuracy.	Oversampling increases data noise with overfitting risks
4	Buraimoh [124]	2021	student success	500 / 16	LMS Behavioral Log Data, Demographic and Academic	Logistic Regression, SVM, Naïve Bayes, DT, Random Forests, GBT, MLP, and LDA	Deep Learning	• Student absence days, raised hands, visited resources, announcement view, and parent answering a survey are the top features. • Random Forest surpassed other models with an accuracy rate of **90%.**	Small dataset; Oversampling increase data noise with overfitting risks
5	Madnaik [126]	2020	student's grade	480 / 17; 395 / 31	Social factors (parents' job and education, alcohol consumption, and student's health); Demographic, Academic background & LMS Data	Random Forest, K-Nearest Neighbors, Gaussian Naïve Bayes, Bernoulli Naïve Bayes	Supervised ML	• Student grades are mostly affected by student class participation • Random Forest is the best classifier (**93%**)	Doesn't address data imbalance problems.
6	Pojon [128]	2017	student's grade	480 / 17; 395 / 31	Social factors (parents' job and education, alcohol consumption, and student's health); Demographic, Academic & LMS Data	Linear regression, Decision trees & Naïve Bayes	Supervised ML	• Feature engineering improves the accuracy of models • Naïve Bayes (**97.5%**)	Limited Sample size
7	Mitra & Le [81]	2019	student performance	128 / 18	cognitive and behavioral variables	Quantile regression, LR, NB, SVM, DT, RF	Analysis and Statistical approaches & Supervised ML	• Resulted in the predictive accuracy of over **95%**	Limited sample size; lack of general inferences
8	Umer et al. [106]	2019	at-risk student	125 / N.A.	Moodle interactions and scores	RF, LDA, NB, LR & ensemble classifier	Supervised ML	• Combining features from LMS and assessment grades helps to increase prediction accuracy.	It May not be generalizable
9	Fahd et al. [76]	2021	at-risk student	112 / 13	LMS Interaction Logs	Random forest, J48, NBTree, OneR, and Decision stump	Supervised ML	• Random forest is the best classifier with an **85.7%** accuracy rate.	Does not consider behavioral factors; limited sample size
10	Glavas et al. [100]	2018	student's grades	77 / 17	Academic, forums, LMS interactions	Ridge regression & Lasso regression	Analysis and Statistical approaches	• Lasso regression performed the best to eliminate less significant predictors.	May not work with numeric attributes
11	Hasan [101]	2018	prediction of student performance	22 / N.A.	Student's Academic and Students Activity (On and off Campus)	J48, REP Tree, RF, Logistic Model Tree, SMO, Hoeffding Tree, Decision Stump, NB	Supervised ML	• SMO outperforms other classifiers with an accuracy rate of **95.2%**.	Limited Sample size
12	Macarini [79]	2019	Student Dropout	N.A.	Moodle Logs: Interaction Count of Cognitive, Teaching, Social and Others	AdaBoost, Random Forest, KNN, Naïve Bayes, MLP	Deep Learning	• Questionnaire data does not improve prediction; unless feature selection is performed.	Oversampling increases data noise with overfitting risks

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student-teacher relations, and the qualifications of the assigned teacher may have the potential to boost the prediction accuracy.

6) ***Several ethical issues exist*** there is no doubt that institutions of higher and secondary education have always gathered and analyzed huge volumes of student data, but with the rise in demand for the exploitation of these data, ethical concerns can arise. For example, the collection of the data, the transparency in terms of the purpose of gathering such sensitive data, the privacy, and security, and informed consent. As observed from this review, few researchers addressed these issues for example through signed agreements regarding the disclosure of the sensitive data, verifying the unique student identifiers and anonymizing student's names, and giving out consent forms to students to sign. The upcoming research should acknowledge the ethical aspect of their predictive models by including the measures taken to ensure the privacy of the user data.
7) ***Lack of unsupervised learning techniques*** such as clustering to forecast student performance. It is concluded that most researchers utilized supervised and

deep learning models, however, models exploring other techniques such as clustering to understand the behaviour of students are generally lacking. This is another valuable challenge that can be addressed by further researchers on this topic. For example, segregating students based on their observable and similar patterns to provide feedback tailored for each sub-population of a group of students. Such an approach will emphasize that students are not a homogeneous (same characteristics) group: classify some students as at-risk because they are academically weak, others predict as dropouts

VI. CONCLUSION

In institutions, it is critical to evaluate student performance as early as possible. Overall, this systematic review has achieved the intended results of identifying the current Learning Analytics approaches and algorithms to enhance education and student's learning style. The study reveals that supervised Machine Learning is among the most applied approach for predictive models in student retention (53 studies). Interestingly, nearly 47% of researchers focused on a student-centred approach, particularly for undergraduate students (50 studies). Moreover, it has been found that Asia is among the leading continent in this field (27 studies). However, the study suggests that the application of LA and EDM is on the rise to build more robust predictive models for student retention.

A. RECOMMENDATIONS

The upcoming future research will need to focus on exploring more ensemble and clustering methods to efficiently predict the performance of students and enhance the prediction accuracy. Additionally, other dynamic and social factors can be explored to further understand student behaviour as suggested by this review, following refined data pre-processing techniques. Lastly, privacy and ethical concerns should be addressed more comprehensibly by future researchers. We suggest that more careful procedures to be taken pertaining to the transparency of data collection, the ownership and storage of data, and finally, the required consent from stakeholders. The findings concerning the ethical constraints align with other empirical and review studies found in [16], [31], [142].

B. LIMITATIONS

Our study is limited by its scope of reviewing articles in the EDM and LA field published in the past five years only. Moreover, as this is a systematic literature review, we believe that there are many areas to explore in the presented articles to conclude a more theoretical contribution rather than methodological implications, which is a limitation of this review.

C. FUTURE DIRECTIONS

Lastly, as a future work for this review, more analysis of articles and comparisons will be made to compare the results of these predictive models in terms of accuracy and chosen methods to draw out specific theoretical conclusions for each type of learning environment. Moreover, this review can be extended further by explaining the application of LA to build assertive remedial plans and choosing the appropriate teaching methods.

APPENDIX

D. DATA SOURCES (FEATURES) USED IN LEARNING ANALYTICS

See Table 9.

E. COMPARATIVE SUMMARY OF THE SURVEYED ARTICLES

See Tables 10–12.

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REFERENCES

[1] G. D. Caruth, "Student engagement, retention, and motivation: Assessing academic success in today's college students," *Participatory Educ. Res.*, vol. 5, no. 1, pp. 17–30, Dec. 2018, doi: 10.17275/per.18.4.5.1.

[2] L. S. Hagedorn, "How to define retention: A new look at an old problem," USC Rossier School Educ., Los Angeles, CA, USA, Tech. Rep., 2006.

[3] M. Haverila, K. Haverila, and C. McLaughlin, "Variables affecting the retention intentions of students in higher education institutions," *J. Int. Students*, vol. 10, no. 2, pp. 358–382, May 2020, doi: 10.32674/jis.v10i2.1849.

[4] C. S. Sarrico, "Completion and retention in higher education," in *Encyclopedia of International Higher Education Systems and Institutions*. Dordrecht, The Netherlands: Springer, 2018, pp. 1–5.

[5] B. Albreiki, N. Zaki, and H. Alashwal, "A systematic literature review of student' performance prediction using machine learning techniques," *Educ. Sci.*, vol. 11, no. 9, p. 552, Sep. 2021, doi: 10.3390/educsci11090552.

[6] N. M. Alsharari and M. T. Alshurideh, "Student retention in higher education: The role of creativity, emotional intelligence and learner autonomy," *Int. J. Educ. Manage.*, vol. 35, no. 1, pp. 233–247, Jan. 2021, doi: 10.1108/IJEM-12-2019-0421.

[7] D. Kucak, V. Juricic, and G. Dambic, "Machine learning in education—A survey of current research trends," in *Proc. Ann. DAAAM Int. DAAAM Symp.*, 2018, vol. 29, no. 1, pp. 406–410.

[8] C. Miller, "College dropout rates," Educ. Data, educationdata.org, USA, Tech. Rep., 2019.

[9] M. Hanson. (Sep. 14, 2021). *College Dropout Rate [2021]: By Year + Demographics*. Accessed: Sep. 15, 2021. [Online]. Available: https://educationdata.org/college-dropout-rates

[10] J. M. Ortiz-Lozano, A. Rua-Vieites, P. Bilbao-Calabuig, and M. Casadesús-Fa, "University student retention: Best time and data to identify undergraduate students at risk of dropout," *Innov. Educ. Teach. Int.*, vol. 57, no. 1, pp. 74–85, 2020, doi: 10.1080/14703297.2018.1502090.

[11] I. S. Pratt, H. B. Harwood, J. T. Cavazos, and C. P. Ditzfeld, "Should i stay or should i go? Retention in first-generation college students," *J. College Student Retention, Res., Theory Pract.*, vol. 21, no. 1, pp. 105–118, May 2019, doi: 10.1177/1521025117690868.

[12] A. Behr, M. Giese, H. D. Teguim Kamdjou, and K. Theune, "Dropping out of university: A literature review," *Rev. Educ.*, vol. 8, no. 2, pp. 614–652, Jun. 2020, doi: 10.1002/rev3.3202.

[13] N. S. Sani, A. Fikri, Z. Ali, M. Zakree, and K. Nadiyah, "Dropout prediction in higher education among B40 students," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 11, pp. 550–559, 2020, doi: 10.14569/IJACSA.2020.0111169.

[14] Z. Shuqfa and S. Harous, "Data mining techniques used in predicting Student retention in higher education: A survey," in *Proc. Int. Conf. Electr. Comput. Technol. Appl. (ICECTA)*, Nov. 2019, pp. 1–4, doi: 10.1109/ICECTA48151.2019.8959789.

[15] O. H. T. Lu, A. Y. Q. Huang, J. C. H. Huang, A. J. Q. Lin, H. Ogata, and S. J. H. Yang, "Applying learning analytics for the early prediction of students' academic performance in blended learning," *J. Educ. Technol. Soc.*, vol. 21, no. 2, pp. 220–232, 2018.

[16] A. Khan, S. K. Ghosh, D. Ghosh, and S. Chattopadhyay, "Random wheel: An algorithm for early classification of student performance with confidence," *Eng. Appl. Artif. Intell.*, vol. 102, Jun. 2021, Art. no. 104270, doi: 10.1016/j.engappai.2021.104270.

[17] O. Viberg, M. Hatakka, O. Bälter, and A. Mavroudi, "The current landscape of learning analytics in higher education," *Comput. Hum. Behav.*, vol. 89, pp. 98–110, Dec. 2018, doi: 10.1016/j.chb.2018.07.027.

[18] B. Tak and M. Wong, "Learning analytics in higher education: An analysis of case studies," *Asian Assoc. Open Univ. J.*, vol. 12, no. 1, pp. 21–40, May 2017, doi: 10.1108/AAOUJ-01-2017-0009.

[19] J. Barshay and S. Aslanian, "Colleges are using big data to track students in an effort to boost graduation rates, but it comes at a cost," APM Rep., MN, USA, Tech. Rep., 2019.

[20] A. S. Hashim, W. A. Awadh, and A. K. Hamoud, "Student performance prediction model based on supervised machine learning algorithms," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 928, no. 3, Nov. 2020, Art. no. 032019, doi: 10.1088/1757-899X/928/3/032019.

[21] B. Bakhshinategh, O. R. Zaiane, S. ElAtia, and D. Ipperciel, "Educational data mining applications and tasks: A survey of the last 10 years," *Educ. Inf. Technol.*, vol. 23, no. 1, pp. 537–553, Jan. 2018, doi: 10.1007/s10639-017-9616-z.

[22] U. Fayyad, G. Piatetsky-Shapiro, and P. Smyth, "The KDD process for extracting useful knowledge from volumes of data," *Commun. ACM*, vol. 39, no. 11, pp. 27–34, Nov. 1996, doi: 10.1145/240455.240464.

[23] E. Sultan, M. Norshahriel, A. Rani, N. Filzah, M. Radzuan, and L. Huay, "Predictive analytics on university student dropouts from online learning due to MCO," in *Proc. Knowl. Manage. Int. Conf. (KMICe)*, Feb. 2021, pp. 117–123.

[24] S. Shrestha and M. Pokharel, "Data mining applications used in education sector," *J. Educ. Res.*, vol. 10, no. 2, pp. 27–51, Nov. 2020, doi: 10.3126/jer.v10i2.32721.

[25] D. West, D. Heath, and H. Huijser, "Let's talk learning analytics: A framework for implementation in relation to student retention," *J. Asynchronous Learn. Netw.*, vol. 20, no. 2, pp. 30–50, 2016, doi: 10.24059/olj.v20i2.792.

[26] *The Possibilities of Using Learning Analytics in Higher Education*, Talismatic, Princeton, NJ, USA, 2018.

[27] A. A. Mubarak, H. Cao, and I. M. Hezam, "Deep analytic model for student dropout prediction in massive open online courses," *Comput. Electr. Eng.*, vol. 93, Jul. 2021, Art. no. 107271, doi: 10.1016/j.compeleceng.2021.107271.

[28] D. J. Lemay, C. Baek, and T. Doleck, "Comparison of learning analytics and educational data mining: A topic modeling approach," *Comput. Educ., Artif. Intell.*, vol. 2, 2021, Art. no. 100016, doi: 10.1016/j.caeai.2021.100016.

[29] M. Şahin and H. Yurdugül, "Educational Data Mining and Learning Analytics: Past, present and future," *Bartın Univ. J. Fac. Educ.*, vol. 9, no. 1, pp. 121–131, 2019. [Online]. Available: https://www.semanticscholar.org/paper/Educational-Data-Mining-and-Learning-Analytics%3A-and-%C5%9Eahin-Yurdug%C3%BCl/c88bb0131bc1d346352330cbce3a0f6a5a80b0fd

[30] C. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," *WIREs Data Mining Knowl. Discovery*, vol. 10, no. 3, pp. 1–21, May 2020, doi: 10.1002/widm.1355.

[31] S. Dormezil, T. Khoshgoftaar, and F. Robinson-Bryant, "Differentiating between educational data mining and learning analytics: A bibliometric approach," in *Proc. CEUR Workshop*, vol. 2592, 2020, pp. 17–22.

[32] M. Liz-Domínguez, M. Caeiro-Rodríguez, M. Llamas-Nistal, and F. A. Mikic-Fonte, "Systematic literature review of predictive analysis tools in higher education," *Appl. Sci.*, vol. 9, no. 24, p. 5569, Dec. 2019, doi: 10.3390/app9245569.

[33] M. Kumar, A. J. Singh, and D. Handa, "Literature survey on student's performance prediction in education using data mining techniques," *Int. J. Educ. Manage. Eng.*, vol. 7, no. 6, pp. 40–49, Nov. 2017, doi: 10.5815/ijeme.2017.06.05.

[34] X. Hu, C. W. L. Cheong, W. Ding, and M. Woo, "A systematic review of studies on predicting Student learning outcomes using learning analytics," in *Proc. 7th Int. Learn. Anal. Knowl. Conf.*, Mar. 2017, pp. 528–529, doi: 10.1145/3027385.3029438.

[35] A. Hellas, P. Ihantola, A. Petersen, V. V. Ajanovski, M. Gutica, T. Hynninen, A. Knutas, J. Leinonen, C. Messom, and S. N. Liao, "Predicting academic performance: A systematic literature review," in *Proc. Companion 23rd Annu. ACM Conf. Innov. Technol. Comput. Sci. Educ.*, Jul. 2018, pp. 175–199, doi: 10.1145/3293881.3295783.

[36] F. Dalipi, A. S. Imran, and Z. Kastrati, "MOOC dropout prediction using machine learning techniques: Review and research challenges," in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, Apr. 2018, pp. 1007–1014, doi: 10.1109/EDUCON.2018.8363340.

[37] A. Mat, N. Z. Khidzir, S. Deris, N. N. H. Mat Yaacob, M. S. Mohamad, and S. S. Safaai, "A review on predictive modeling technique for student academic performance monitoring," in *Proc. MATEC Web Conf.*, vol. 255, 2019, p. 3004, doi: 10.1051/matecconf/201925503004.

[38] J. L. Rastrollo-Guerrero, J. A. Gómez-Pulido, and A. Durán-Domínguez, "Analyzing and predicting students' performance by means of machine learning: A review," *Appl. Sci.*, vol. 10, no. 3, p. 1042, Feb. 2020, doi: 10.3390/app10031042.

[39] E. Alyahyan and D. Düştegör, "Predicting academic success in higher education: Literature review and best practices," *Int. J. Educ. Technol. Higher Educ.*, vol. 17, no. 1, pp. 1–21, Dec. 2020, doi: 10.1186/s41239-020-0177-7.

[40] A. Namoun and A. Alshanqiti, "Predicting student performance using data mining and learning analytics techniques: A systematic literature review," *Appl. Sci.*, vol. 11, no. 1, p. 237, Dec. 2020, doi: 10.3390/app11010237.

[41] G. M. Tawfik, K. A. S. Dila, M. Y. F. Mohamed, D. N. H. Tam, N. D. Kien, A. M. Ahmed, and N. T. Huy, "A step by step guide for conducting a systematic review and meta-analysis with simulation data," *Trop. Med. Health*, vol. 47, no. 1, p. 46, Dec. 2019, doi: 10.1186/s41182-019-0165-6.

[42] P. Ryś, M. Władysiuk, I. Skrzekowska-Baran, and M. T. Małecki, "Review articles, systematic reviews and meta-analyses: Which can be trusted?" *Polish Arch. Internal Med.*, vol. 119, no. 3, pp. 148–156, Mar. 2009, doi: 10.20452/pamw.634.

[43] M. J. Page *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *Int. J. Surg.*, vol. 88, Apr. 2021, Art. no. 105906, doi: 10.1016/j.ijsu.2021.105906.

[44] S. Priya, T. Ankit, and D. Divyansh, "Student performance prediction using machine learning," in *Advances in Parallel Computing Technologies and Applications*, vol. 39. Amsterdam, The Netherlands: IOS Press, 2021, pp. 167–174, doi: 10.3233/APC210137.

[45] S. Hussain and M. Q. Khan, "Student-performulator: Predicting students' academic performance at secondary and intermediate level using machine learning," *Ann. Data Sci.*, pp. 1–19, Jun. 2021, doi: 10.1007/s40745-021-00341-0.

[46] A. Marwaha and A. Singla, "A study of factors to predict at-risk students based on machine learning techniques," in *Intelligent Communication, Control and Devices* (Advances in Intelligent Systems and Computing), vol. 989. Singapore: Springer, Jan. 2020, pp. 133–141, doi: 10.1007/978-981-13-8618-3_15.

[47] S. Bundasak, "Student behavior analysis affecting learning achievement of information technology and computer science students," *Int. J. Mach. Learn. Comput.*, vol. 10, no. 2, pp. 346–351, Feb. 2020, doi: 10.18178/ijmlc.2020.10.2.941.

[48] V. Arhin and J. E. Laryea, "Tutoring support as a predictor of student retention in distance learning: The case of a University in Ghana," *Open Praxis*, vol. 12, no. 4, pp. 457–468, 2020, doi: 10.5944/openpraxis.12.4.1124.

[49] H. Altabrawee, O. A. J. Ali, and S. Q. Ajmi, "Predicting students' performance using machine learning techniques," *J. Univ. BABYLON Pure Appl. Sci.*, vol. 27, no. 1, pp. 194–205, Apr. 2019, doi: 10.29196/jubpas.v27i1.2108.

[50] C. C. Gray and D. Perkins, "Utilizing early engagement and machine learning to predict student outcomes," *Comput. Educ.*, vol. 131, pp. 22–32, Apr. 2019, doi: 10.1016/j.compedu.2018.12.006.

[51] S. Lee and J. Y. Chung, "The machine learning-based dropout early warning system for improving the performance of dropout prediction," *Appl. Sci.*, vol. 9, no. 15, p. 3093, Jul. 2019, doi: 10.3390/app9153093.

[52] A. K. Hamoud, A. S. Hashim, and W. A. Awadh, "Predicting Student performance in higher education institutions using decision tree analysis," *Int. J. Interact. Multimedia Artif. Intell.*, vol. 5, no. 2, p. 26, 2018, doi: 10.9781/ijimai.2018.02.004.

[53] A. Soni, V. Kumar, R. Kaur, and D. Hemavathi, "Predicting student performance using data mining techniques," *Int. J. Pure Appl. Math.*, vol. 119, no. 12, pp. 221–227, Oct. 2018, doi: 10.26438/ijcse/v6i10.172177.

[54] S. V. Parmar and L. K. Sharma, "Comparative study of supervised learning for student performance evaluation," *Int. J. Comput. Eng. Technol.*, vol. 9, no. 2, pp. 1–8, 2018.

[55] B. K. Yousafzai, S. A. Khan, T. Rahman, I. Khan, I. Ullah, A. U. Rehman, M. Baz, H. Hamam, and O. Cheikhrouhou, "Student-performulator: Student academic performance using hybrid deep neural network," *Sustainability*, vol. 13, no. 17, p. 9775, Aug. 2021, doi: 10.3390/su13179775.

[56] S. Rovira, E. Puertas, and L. Igual, "Data-driven system to predict academic grades and dropout," *PLoS ONE*, vol. 12, no. 2, Feb. 2017, Art. no. e0171207, doi: 10.1371/journal.pone.0171207.

[57] G. M. Rao and P. K. K. Kumar, "Students performance prediction in online courses using machine learning algorithms," *United Int. J. Res. Technol.*, vol. 2, no. 11, pp. 74–79, 2021.

[58] M. Adnan, A. Habib, J. Ashraf, S. Mussadiq, A. A. Raza, M. Abid, M. Bashir, and S. U. Khan, "Predicting at-risk students at different percentages of course length for early intervention using machine learning models," *IEEE Access*, vol. 9, pp. 7519–7539, 2021, doi: 10.1109/ACCESS.2021.3049446.

[59] H. Waheed, S.-U. Hassan, N. R. Aljohani, J. Hardman, S. Alelyani, and R. Nawaz, "Predicting academic performance of students from VLE big data using deep learning models," *Comput. Hum. Behav.*, vol. 104, Mar. 2020, Art. no. 106189, doi: 10.1016/j.chb.2019.106189.

[60] D. Bañeres, M. E. Rodríguez, A. E. Guerrero-Roldán, and A. Karadeniz, "An early warning system to detect at-risk students in online higher education," *Appl. Sci.*, vol. 10, no. 13, p. 4427, Jun. 2020, doi: 10.3390/app10134427.

[61] V. S. Vamshidharreddy, A. S. Saketh, and R. Gnanajeyaraman, "Student's academic performance prediction using machine learning approach," *Int. J. Adv. Sci. Technol.*, vol. 29, no. 9, pp. 6731–6737, 2020.

[62] R. Hasan, S. Palaniappan, S. Mahmood, A. Abbas, K. U. Sarker, and M. U. Sattar, "Predicting student performance in higher educational institutions using video learning analytics and data mining techniques," *Appl. Sci.*, vol. 10, no. 11, p. 3894, Jun. 2020, doi: 10.3390/app10113894.

[63] A. Y. Q. Huang, O. H. T. Lu, J. C. H. Huang, C. J. Yin, and S. J. H. Yang, "Predicting students' academic performance by using educational big data and learning analytics: Evaluation of classification methods and learning logs," *Interact. Learn. Environ.*, vol. 28, no. 2, pp. 206–230, Feb. 2020, doi: 10.1080/10494820.2019.1636086.

[64] Y. Cui, F. Chen, and A. Shiri, "Scale up predictive models for early detection of at-risk students: A feasibility study," *Inf. Learn. Sci.*, vol. 121, nos. 3–4, pp. 97–116, Feb. 2020, doi: 10.1108/ILS-05-2019-0041.

[65] S. Rizvi, B. Rienties, and S. A. Khoja, "The role of demographics in online learning; a decision tree based approach," *Comput. Educ.*, vol. 137, pp. 32–47, Aug. 2019, doi: 10.1016/j.compedu.2019.04.001.

[66] T. Soffer and A. Cohen, "Students' engagement characteristics predict success and completion of online courses," *J. Comput. Assist. Learn.*, vol. 35, no. 3, pp. 378–389, Jun. 2019, doi: 10.1111/jcal.12340.

[67] M. Akour, H. A. Sghaier, and O. A. Qasem, "The effectiveness of using deep learning algorithms in predicting students achievements," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 19, no. 1, pp. 388–394, Jul. 2020, doi: 10.11591/ijeecs.v19.i1.pp388-394.

[68] N. Nadar and R. Kamatchi, "A novel student risk identification model using machine learning approach," *Int. J. Adv. Comput. Sci. Appl.*, vol. 9, no. 11, pp. 305–309, 2018, doi: 10.14569/ijacsa.2018.091142.

[69] J. Bainbridge, J. Melitski, A. Zahradnik, E. J. M. Lauría, S. Jayaprakash, and J. Baron, "Using learning analytics to predict at-risk students in online graduate public affairs and administration education," *J. Public Affairs Educ.*, vol. 21, no. 2, pp. 247–262, Jun. 2015, doi: 10.1080/15236803.2015.12001831.

[70] M. Hussain, W. Zhu, W. Zhang, and S. M. R. Abidi, "Student engagement predictions in an e-Learning system and their impact on Student course assessment scores," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–21, Oct. 2018, doi: 10.1155/2018/6347186.

[71] S. Helal, J. Li, L. Liu, E. Ebrahimie, S. Dawson, D. J. Murray, and Q. Long, "Predicting academic performance by considering student heterogeneity," *Knowl.-Based Syst.*, vol. 161, pp. 134–146, Dec. 2018, doi: 10.1016/j.knosys.2018.07.042.

[72] P. P. Bendangnuksung and D. Prabu, "Students' performance prediction using deep neural network," *Int. J. Appl. Eng. Res.*, vol. 13, no. 2, pp. 1171–1176, 2018.

[73] B. E. Shelton, J.-L. Hung, and P. R. Lowenthal, "Predicting student success by modeling student interaction in asynchronous online courses," *Distance Educ.*, vol. 38, no. 1, pp. 59–69, Jan. 2017, doi: 10.1080/01587919.2017.1299562.

[74] R. Umer, T. Susnjak, A. Mathrani, and S. Suriadi, "On predicting academic performance with process mining in learning analytics," *J. Res. Innov. Teaching Learn.*, vol. 10, no. 2, pp. 160–176, Jul. 2017, doi: 10.1108/JRIT-09-2017-0022.

[75] K. Ahammad, P. Chakraborty, E. Akter, U. H. Fomey, and S. Rahman, "A comparative study of different machine learning techniques to predict the result of an individual student using previous performances," *Int. J. Comput. Sci. Inf. Secur.*, vol. 19, no. 1, pp. 5–10, 2021, doi: 10.5281/zenodo.4533374.

[76] K. Fahd, S. J. Miah, and K. Ahmed, "Predicting student performance in a blended learning environment using learning management system interaction data," *Appl. Comput. Informat.*, Oct. 2021. [Online]. Available: https://www.emerald.com/insight/content/doi/10.1108/ACI-06-2021-0150/full/html, doi: 10.1108/ACI-06-2021-0150.

[77] M. N. Yakubu and A. M. Abubakar, "Applying machine learning approach to predict students' performance in higher educational institutions," *Kybernetes*, vol. 51, no. 2, pp. 916–934, Feb. 2022, doi: 10.1108/K-12-2020-0865.

[78] J. Kabathova and M. Drlik, "Towards predicting student's dropout in university courses using different machine learning techniques," *Appl. Sci.*, vol. 11, no. 7, p. 3130, Apr. 2021, doi: 10.3390/app11073130.

[79] L. A. B. Macarini, C. Cechinel, M. F. B. Machado, V. F. C. Ramos, and R. Munoz, "Predicting students success in blended learning—Evaluating different interactions inside learning management systems," *Appl. Sci.*, vol. 9, no. 24, p. 5523, Dec. 2019, doi: 10.3390/app9245523.

[80] A. Jokhan, B. Sharma, and S. Singh, "Early warning system as a predictor for student performance in higher education blended courses," *Stud. Higher Educ.*, vol. 44, no. 11, pp. 1900–1911, Nov. 2019, doi: 10.1080/03075079.2018.1466872.

[81] S. Mitra and K. Le, "The effect of cognitive and behavioral factors on student success in a bottleneck business statistics course via deeper analytics," *Commun. Stat. Simul. Comput.*, vol. 51, no. 6, pp. 2779–2808, 2022, doi: 10.1080/03610918.2019.1700279.

[82] N. Z. Zacharis, "Classification and regression trees (CART) for predictive modeling in blended learning," *Int. J. Intell. Syst. Appl.*, vol. 10, no. 3, pp. 1–9, Mar. 2018, doi: 10.5815/ijisa.2018.03.01.

[83] M. Saqr, U. Fors, and M. Tedre, "How learning analytics can early predict under-achieving students in a blended medical education course," *Med. Teacher*, vol. 39, no. 7, pp. 757–767, Jul. 2017, doi: 10.1080/0142159X.2017.1309376.

[84] R. R. Estacio and R. C. Raga, Jr., "Analyzing students online learning behavior in blended courses using moodle," *Asian Assoc. Open Univ. J.*, vol. 12, no. 1, pp. 52–68, May 2017, doi: 10.1108/AAOUJ-01-2017-0016.

[85] K. Maheswari, A. Priya, A. Balamurugan, and S. Ramkumar, "Analyzing student performance factors using KNN algorithm," *Mater. Today, Proc.*, Feb. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214785320407424?via%3Dihub, doi: 10.1016/j.matpr.2020.12.1024.

[86] O. W. Adejo and T. Connolly, "Predicting student academic performance using multi-model heterogeneous ensemble approach," *J. Appl. Res. Higher Educ.*, vol. 10, no. 1, pp. 61–75, Feb. 2018, doi: 10.1108/JARHE-09-2017-0113.

[87] G. Akçapınar, A. Altun, and P. Aşkar, "Using learning analytics to develop early-warning system for at-risk students," *Int. J. Educ. Technol. Higher Educ.*, vol. 16, no. 1, p. 40, Dec. 2019, doi: 10.1186/s41239-019-0172-z.

[88] J. Bravo-Agapito, S. J. Romero, and S. Pamplona, "Early prediction of undergraduate student's academic performance in completely online learning: A five-year study," *Comput. Hum. Behav.*, vol. 115, Feb. 2021, Art. no. 106595, doi: 10.1016/j.chb.2020.106595.

[89] K. T. Chui, D. C. L. Fung, M. D. Lytras, and T. M. Lam, "Predicting at-risk university students in a virtual learning environment via a machine learning algorithm," *Comput. Hum. Behav.*, vol. 107, Jun. 2020, Art. no. 105584, doi: 10.1016/j.chb.2018.06.032.

[90] Y. Chen, Q. Zheng, S. Ji, F. Tian, H. Zhu, and M. Liu, "Identifying at-risk students based on the phased prediction model," *Knowl. Inf. Syst.*, vol. 62, no. 3, pp. 987–1003, 2020, doi: 10.1007/s10115-019-01374-x.

[91] S. F. A. Aziz, "Students' performance evaluation using machine learning algorithms," *College Basic Educ. Res. J.*, vol. 16, no. 3, pp. 976–986, 2020.

[92] L. Kemper, G. Vorhoff, and B. U. Wigger, "Predicting student dropout: A machine learning approach," *Eur. J. Higher Educ.*, vol. 10, no. 1, pp. 28–47, Jan. 2020, doi: 10.1080/21568235.2020.1718520.
[93] J. S. Gil, A. J. P. Delima, and R. N. Vilchez, "Predicting students' dropout indicators in public school using data mining approaches," *Int. J. Adv. Trends Comput. Sci. Eng.*, vol. 9, no. 1, pp. 774–778, Feb. 2020.
[94] I. Khan, A. A. Sadiri, A. R. Ahmad, and N. Jabeur, "Tracking student performance in introductory programming by means of machine learning," in *Proc. 4th MEC Int. Conf. Big Data Smart City (ICBDSC)*, Jan. 2019, pp. 1–6, doi: 10.1109/ICBDSC.2019.8645608.
[95] J. Nouri, K. Larsson, and M. Saqr, "Identifying factors for master thesis completion and non-completion through learning analytics and machine learning," in *Proc. 14th Eur. Conf. Technol. Enhanced Learn.*, vol. 9, 2019, pp. 28–39, doi: 10.1007/978-3-030-29736-7_3.
[96] A. Polyzou and G. Karypis, "Feature extraction for classifying students based on their academic performance," in *Proc. 11th Int. Conf. Educ. Data Mining (EDM)*, 2018, pp. 356–362.
[97] H. Chanlekha and J. Niramitranon, "Student performance prediction model for early-identification of at-risk students in traditional classroom settings," in *Proc. 10th Int. Conf. Manage. Digit. EcoSyst.*, Sep. 2018, pp. 239–245, doi: 10.1145/3281375.3281403.
[98] F. J. Kaunang and R. Rotikan, "Students' academic performance prediction using data mining," in *Proc. 3rd Int. Conf. Informat. Comput. (ICIC)*, Oct. 2018, pp. 1–5, doi: 10.1109/IAC.2018.8780547.
[99] G.-H. Wang, J. Zhang, and G.-S. Fu, "Predicting Student behaviors and performance in online learning using decision tree," in *Proc. 7th Int. Conf. Educ. Innov. Technol. (EITT)*, Dec. 2018, pp. 214–219, doi: 10.1109/EITT.2018.00050.
[100] M. Glavas, M. B. Bakaric, and M. Matetic, "Applying advanced linear models in the task of predicting Student success," in *Proc. 41st Int. Conv. Inf. Commun. Technol., Electron. Microelectron. (MIPRO)*, May 2018, pp. 744–748, doi: 10.23919/MIPRO.2018.8400138.
[101] R. Hasan, S. Palaniappan, A. R. A. Raziff, S. Mahmood, and K. U. Sarker, "Student academic performance prediction by using decision tree algorithm," in *Proc. 4th Int. Conf. Comput. Inf. Sci. (ICCOINS)*, Aug. 2018, pp. 1–5, doi: 10.1109/ICCOINS.2018.8510600.
[102] L. C. Ho and K. J. Shim, "Data mining approach to the identification of at-risk students," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2018, pp. 5333–5335, doi: 10.1109/BigData.2018.8622495.
[103] W. Han, D. Jun, L. Kangxu, and G. Xiaopeng, "Investigating performance in a blended SPOC," in *Proc. IEEE 6th Int. Conf. Teaching, Assessment, Learn. Eng. (TALE)*, Dec. 2017, pp. 239–245, doi: 10.1109/TALE.2017.8252340.
[104] J. Malini and Y. Kalpana, "Investigation of factors affecting student performance evaluation using education materials data mining technique," *Mater. Today, Proc.*, vol. 47, pp. 6105–6110, Jan. 2021, doi: 10.1016/j.matpr.2021.05.026.
[105] N. Kondo, M. Okubo, and T. Hatanaka, "Early detection of at-risk students using machine learning based on LMS log data," in *Proc. 6th IIAI Int. Congr. Adv. Appl. Informat. (IIAI-AAI)*, Jul. 2017, pp. 198–201, doi: 10.1109/IIAI-AAI.2017.51.
[106] R. Umer, A. Mathrani, T. Susnjak, and S. Lim, "Mining activity log data to predict student's outcome in a course," in *Proc. Int. Conf. Big Data Educ. (ICBDE)*, 2019, pp. 52–58, doi: 10.1145/3322134.3322140.
[107] M. Nagy and R. Molontay, "Predicting dropout in higher education based on secondary school performance," in *Proc. IEEE 22nd Int. Conf. Intell. Eng. Syst. (INES)*, Jun. 2018, pp. 389–394, doi: 10.1109/INES.2018.8523888.
[108] S. S. Athani, S. A. Kodli, M. N. Banavasi, and P. G. S. Hiremath, "Student performance predictor using multiclass support vector classification algorithm," in *Proc. Int. Conf. Signal Process. Commun. (ICSPC)*, Jul. 2017, pp. 341–346, doi: 10.1109/CSPC.2017.8305866.
[109] A. Daud, N. R. Aljohani, R. A. Abbasi, M. D. Lytras, F. Abbas, and J. S. Alowibdi, "Predicting Student performance using advanced learning analytics," in *Proc. 26th Int. Conf. World Wide Web Companion (WWW Companion)*, 2017, pp. 415–421, doi: 10.1145/3041021.3054164.
[110] A. U. Khasanah and Harwati, "A comparative study to predict student's performance using educational data mining techniques," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 215, no. 1, Jun. 2017, Art. no. 012036, doi: 10.1088/1757-899X/215/1/012036.
[111] V. T. Ngoc Chau and N. H. Phung, "A robust random forest-based tri-training algorithm for early in-trouble student prediction," in *Proc. 4th NAFOSTED Conf. Inf. Comput. Sci.*, Nov. 2017, pp. 84–89, doi: 10.1109/NAFOSTED.2017.8108043.
[112] D. S. Jeslet, D. Komarasamy, and J. J. Hermina, "Student result prediction in COVID-19 lockdown using machine learning techniques," *J. Phys., Conf. Ser.*, vol. 1911, no. 1, May 2021, Art. no. 012008, doi: 10.1088/1742-6596/1911/1/012008.
[113] S. D. A. Bujang, A. Selamat, and O. Krejcar, "A predictive analytics model for students grade prediction by supervised machine learning," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 1051, no. 1, Feb. 2021, Art. no. 012005, doi: 10.1088/1757-899X/1051/1/012005.
[114] H. Pallathadka, A. Wenda, E. Ramirez-Asís, M. Asís-López, J. Flores-Albornoz, and K. Phasinam, "Classification and prediction of student performance data using various machine learning algorithms," *Mater. Today, Proc.*, Jul. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S221478532105241X?via%3Dihub, doi: 10.1016/j.matpr.2021.07.382.
[115] P. Dabhade, R. Agarwal, K. P. Alameen, A. T. Fathima, R. Sridharan, and G. Gopakumar, "Educational data mining for predicting students' academic performance using machine learning algorithms," *Mater. Today, Proc.*, vol. 47, pp. 5260–5267, Jan. 2021, doi: 10.1016/j.matpr.2021.05.646.
[116] W. F. W. Yaacob, N. M. Sobri, S. A. M. Nasir, W. F. W. Yaacob, N. D. Norshahidi, and W. Z. W. Husin, "Predicting student drop-out in higher institution using data mining techniques," *J. Phys., Conf. Ser.*, vol. 1496, Mar. 2020, Art. no. 012005, doi: 10.1088/1742-6596/1496/1/012005.
[117] A. K. Veerasamy, D. D'Souza, M.-V. Apiola, M.-J. Laakso, and T. Salakoski, "Using early assessment performance as early warning signs to identify at-risk students in programming courses," in *Proc. IEEE Frontiers Educ. Conf. (FIE)*, Oct. 2020, pp. 1–9, doi: 10.1109/FIE44824.2020.9274277.
[118] N. Chauhan, K. Shah, D. Karn, and P. J. Dala, "Prediction of student performance using machine learning," in *Proc. 2nd Int. Conf. Adv. Sci. Technol. (ICAST)*, 2019, vol. 8, no. 6, pp. 247–255.
[119] A. Rajuladevi, "A machine learning approach to predict first-year student retention rates at University of Nevada, Las Vegas," M.S. thesis, Univ. Nevada, Las Vegas, Las Vegas, NV, USA, 2018.
[120] N. Mduma, "Data driven approach for predicting student dropout in secondary schools," M.S. thesis, Nelson Mandela Afr. Inst. Sci. Technol., Arusha, Tanzania, 2020. [Online]. Available: https://dspace.nm-aist.ac.tz/handle/20.500.12479/898
[121] A. S. S. Albassam, "Detecting students at-risk using learning analytics," M.S. thesis, Univ. Adelaide, Adelaide, SA, Australia, 2019.
[122] S. Iqbal, M. Muntaha, J. I. Natasha, and D. Sakib, "Early grade prediction using profile data," M.S. thesis, BRAC Univ., Dhaka, Bangladesh, Apr. 2020.
[123] A. S. Rayasam, "Predicting at-risk students from disparate sources of institutional data," M.S. thesis, Massachusetts Inst. Technol., Cambridge, MA, USA, 2020. [Online]. Available: https://hdl.handle.net/1721.1/132861
[124] E. F. Buraimoh, "Predicting student success using student engagement in the online component of a blended-learning course," M.S. thesis, School Comput. Sci. Appl. Math., Dept. Sci., Univ. Witwatersrand, Johannesburg, Johannesburg, South Africa, 2021.
[125] J. Feng, "Predicting students' academic performance with decision and neural network," M.S. thesis, Florida Gulf Coast Univ., Fort Myers, FL, USA, 2019.
[126] S. S. Madnaik, "Predicting students' performance by learning analytics," M.S. thesis, Dept. Comput. Sci., San Jose State Univ., San Jose, CA, USA, 2020.
[127] T. Muncie, "Using machine learning models to predict student retention: Building a state-wide early warning system," M.S. thesis, Dept. Educ. Leadership, Morehead State Univ., Morehead, KY, USA, 2020.
[128] M. Pojon, "Using machine learning to predict student performance," M.S. thesis, Univ. Tampere, Finland, 2017.
[129] C. A. Palacios, J. A. Reyes-Suárez, L. A. Bearzotti, V. Leiva, and C. Marchant, "Knowledge discovery for higher education student retention based on data mining: Machine learning algorithms and case study in Chile," *Entropy*, vol. 23, no. 4, pp. 1–23, 2021, doi: 10.3390/e23040485.
[130] V. Heredia-Jimenez, A. Jiménez, M. Ortiz-Rojas, J. I. Marín, P. M. Moreno-Marcos, P. J. Munoz-Merino, and C. D. Kloos, "An early warning dropout model in higher education degree programs: A case study in Ecuador," in *Proc. CEUR Workshop*, vol. 2704, 2020, pp. 58–67.

[131] N. Putpuek, N. Rojanaprasert, K. Atchariyachanvanich, and T. Thamrongthanyawong, "Comparative study of prediction models for final GPA score: A case study of Rajabhat Rajanagarindra University," in *Proc. IEEE/ACIS 17th Int. Conf. Comput. Inf. Sci. (ICIS)*, Jun. 2018, pp. 92–97, doi: 10.1109/ICIS.2018.8466475.

[132] B. Perez, C. Castellanos, and D. Correal, "Applying data mining techniques to predict Student dropout: A case study," in *Proc. IEEE 1st Colombian Conf. Appl. Comput. Intell. (ColCACI)*, May 2018, pp. 1–6, doi: 10.1109/ColCACI.2018.8484847.

[133] R. K. Jena, "Predicting students' learning style using learning analytics: A case study of business management students from India," *Behav. Inf. Technol.*, vol. 37, nos. 10–11, pp. 978–992, Nov. 2018, doi: 10.1080/0144929X.2018.1482369.

[134] E. M. Queiroga, J. L. Lopes, K. Kappel, M. Aguiar, R. M. Araújo, R. Munoz, R. Villarroel, and C. Cechinel, "A learning analytics approach to identify students at risk of dropout: A case study with a technical distance education course," *Appl. Sci.*, vol. 10, no. 11, p. 3998, Jun. 2020, doi: 10.3390/app10113998.

[135] F. Braz, F. Campos, V. Stroele, and M. Dantas, "An early warning model for school dropout: A case study in E-learning class," in *Proc. Anais do XXX Simpósio Brasileiro de Informática na Educação (SBIE)*, Nov. 2019, p. 1441, doi: 10.5753/cbie.sbie.2019.1441.

[136] I. Mwalumbwe and J. S. Mtebe, "Using learning analytics to predict students' performance in moodle learning management system: A case of Mbeya University of Science and Technology," *Electron. J. Inf. Syst. Developing Countries*, vol. 79, no. 1, pp. 1–13, Mar. 2017, doi: 10.1002/j.1681-4835.2017.tb00577.x.

[137] M. S. Viswanathan and D. S. V. Kumar, "Study of students' performance prediction models using machine learning," *Turkish J. Comput. Math. Educ.*, vol. 12, no. 2, pp. 3085–3091, 2021.

[138] L. Wang, "Heterogeneous data and big data analytics," *Autom. Control Inf. Sci.*, vol. 3, no. 1, pp. 8–15, Aug. 2017, doi: 10.12691/acis-3-1-3.

[139] G. Latif, R. Alghazo, M. A. E. Pilotti, and G. B. Brahim, "Identifying 'at-risk' students: An AI-based prediction approach," *Int. J. Comput. Digit. Syst.*, vol. 11, pp. 1–9, Mar. 2022.

[140] D. Sansone, "Beyond early warning indicators: High school dropout and machine learning," *Oxford Bull. Econ. Statist.*, vol. 81, no. 2, pp. 456–485, Apr. 2019, doi: 10.1111/obes.12277.

[141] A. Nagpal and G. Gabrani, "Python for data analytics, scientific and technical applications," in *Proc. Amity Int. Conf. Artif. Intell. (AICAI)*, Feb. 2019, pp. 140–145, doi: 10.1109/AICAI.2019.8701341.

[142] L. Corrin, M. Scheffel, and D. Gašević, "Learning analytics: Pathways to impact," *Australas. J. Educ. Technol.*, vol. 36, no. 6, pp. 1–6, Dec. 2020, doi: 10.14742/ajet.6853.



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